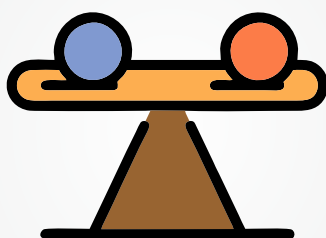


PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

**Center of Mass, Momentum
& Collision**

(English Medium)

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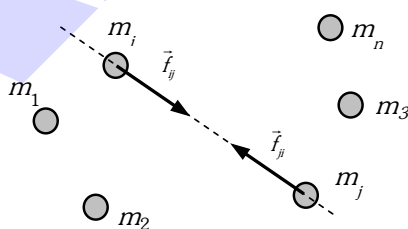
CENTER OF MASS, MOMENTUM & COLLISION

Study of kinematics enables us to explore nature of translation motion without any consideration to forces and energy responsible for the motion. Study of kinetics enables us to explore effects of forces and energy on motion. It includes Newton's laws of motion, methods of work and energy and methods of impulse and momentum. The methods of work and energy and methods of impulse and momentum are developed using equation $\vec{F} = m\vec{a}$ together with the methods of kinematics. The advantage of these methods lie in the fact that they make determination of acceleration unnecessary. Methods of work and energy directly relate force, mass, velocity and displacement and enable us to explore motion between two points of space i.e. in a space interval whereas methods of impulse and momentum enable us to explore motion in a time interval. Moreover methods of impulse and momentum provides only way to analyze impulsive motion.

The work energy theorem and impulse momentum principle are developed from Newton's second law, and we have seen how to apply them to analyze motion of single particle i.e. translation motion of rigid body. Now we will further inquire into possibilities of applying these principles to a system of large number of particles or rigid bodies in translation motion.

1.0 SYSTEM OF PARTICLES

By the term system of particles, we mean a well defined collection of several or large number of particles, which may or may not interact or be connected to each other. As a schematic representation, consider a system of n particles of masses $m_1, m_2, \dots, m_i, \dots, m_j, \dots$ and m_n respectively. They may be actual particles or rigid bodies in translation motion. Some of them may interact with each other and some of them may not. The particles, which interact with each other, apply forces on each other. The forces of interaction $\vec{f}_{ij}$ and $\vec{f}_{ji}$ between a pair of i^{th} and j^{th} particles are shown in the figure. Similar to these other particles may also interact with each other. These forces of mutual interaction between the particles of the system are internal forces of the system.



System of n interacting particles.

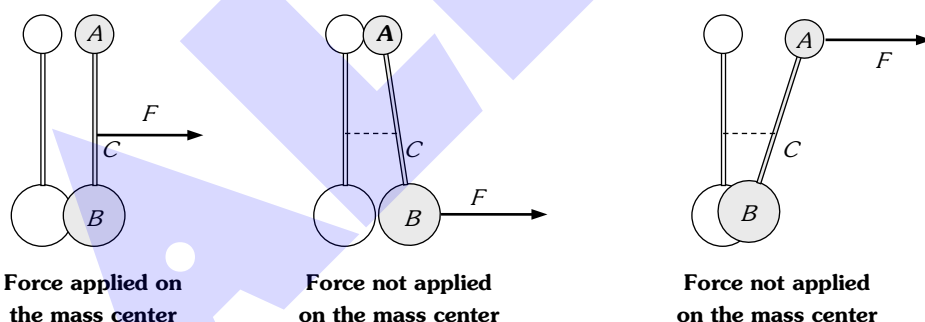
These internal forces always exist in pairs of forces of equal magnitudes and opposite directions. It is not necessary that all the particles interact with each other; some of them, which do not interact with each other, do not apply mutual forces on each other. Other than internal forces, external forces may also act on all or some of the particles. Here by the term external force we mean a force that is applied on any one of the particle included in the system by some other body out-side the system.

In practice we usually deal with extended bodies, which may be deformable or rigid. An extended body is also a system of infinitely large number of particles having infinitely small separations between them. When a body undergoes deformation, separations between its particles and their relative locations change. A rigid body is an extended body in which separations and relative locations of all of its particles remain unchanged under all circumstances.

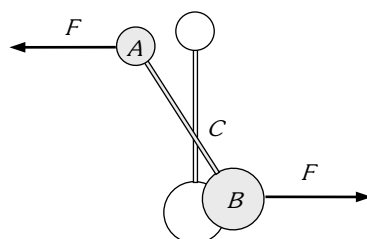
1.1 System of Particles and Mass Center

Until now we have deal with translation motion of rigid bodies, where a rigid body can be treated as a particle. When a rigid body undergoes rotation, all of its particles do not move in identical fashion, still we must treat it a system of particles in which all the particles are rigidly connected to each other. On the other hand we may have particles or bodies not connected rigidly to each other but may be interacting with each other through internal forces. Despite the complex motion of which a system of particles is capable, there is a single point, known as **center of mass** or **mass center** (CM), whose translation motion is characteristic of the system.

The existence of this special point can be demonstrated in the following examples dealing with a rigid body. Consider two disks A and B of unequal masses connected by a very light rigid rod. Place it on a very smooth table. Now pull it horizontally applying a force at different points. You will find a point nearer to the heavier disk, on which if the force is applied the whole assembly undergoes translation motion. Furthermore you cannot find any other point having this property. This point is the mass center of this system. We can assume that all the mass were concentrated at this point. In every rigid body we can find such a point. If you apply the force on any other point, the system moves forward and rotates but the mass center always translates in the direction of the force.

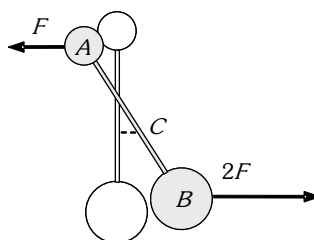


In another experiment, if two forces of equal magnitudes are applied on the disks in opposite directions, the system will rotate, but the mass center C remains stationary as shown in the following figure.



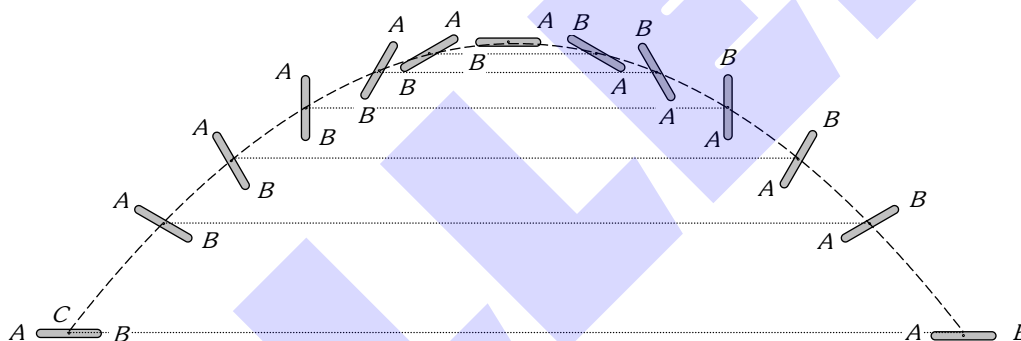
Body rotates but the mass center remains stationary under action of equal and opposite forces.

If the above experiment is repeated with both disks A and B of identical masses, the mass center will be the mid point. And if the experiment is repeated with a uniform rod, the mass center again is the mid point.



Body rotates and the mass center translates under action of unbalanced forces applied at different points.

As another Illustration let us throw a uniform rod in air holding it from one of its ends so that it rotates also. Snapshots taken after regular intervals of time are shown in the figure. The rod rotates through 360° . As the rod moves all of its particles move in a complex manner except the mass center C, which follows a parabolic trajectory as if it were a particle of mass equal to that of the rod and force of gravity were acting on it.



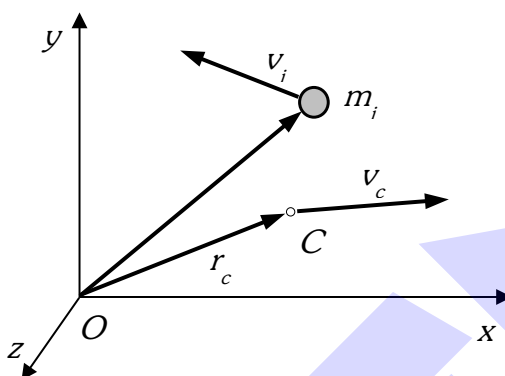
Thus mass center of a rigid body or system of particles is a point, whose translation motion under action of unbalanced forces is same as that of a particle of mass equal to that of the body or system under action the same unbalanced forces. And if different forces having a net resultant are applied at different particles, the system rotates but the mass center translates as if it were a particle of the mass same as that of the system and the net resultant were applied on it.

Concept of mass center provides us a way to look into motion of the system as a whole as superposition of translation of the mass center and motion of all the particles relative to the mass center. In case of rigid bodies all of its particles relative to the mass center can move only on circular paths because they cannot changes their separations.

The concept of mass center is used to represent gross translation of the system. Therefore total linear momentum of the whole system must be equal to the linear momentum of the system due to translation of its mass center.

Center of Mass of System of Discrete Particles

A system of several particles or several bodies having finite separations between them is known as system of discrete particles. Let at an instant particles of such a system $m_1, m_2, \dots, m_i, \dots$ and m_n are moving with velocities $\vec{v}_1, \vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \dots$ and $\vec{v}_n$ at locations $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_i$ and $\vec{r}_n$ respectively. For the sake of simplicity only i^{th} particle and the mass center C are shown in the figure. The mass center C located at $\vec{r}_c$ is moving with velocity $\vec{v}_c$ at this instant.



As the mass center represents gross translation motion of the whole system, the total linear i.e. sum of linear momenta of all the particles must be equal to linear momentum of the whole mass due to translation of the mass center.

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_i \vec{v}_i + \dots + m_n \vec{v}_n = M \vec{v}_c$$

We can write the following equation in terms of masses and position vectors as an analogue to the above equation. This equation on differentiating with respect to time yields the above equation therefore can be thought as solution of the above equation.

$$m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_i \vec{r}_i + \dots + m_n \vec{r}_n = M \vec{r}_c$$

If $M = \sum m_i$ denotes total mass of the system, the above two equations can be written in short as

$$\sum m_i \vec{v}_i = M \vec{v}_c \quad (1)$$

$$\sum m_i \vec{r}_i = M \vec{r}_c \quad (2)$$

The above equation suggests location of mass center of a system of discrete particles.

$$\vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_i \vec{r}_i + \dots + m_n \vec{r}_n}{M} = \frac{\sum m_i \vec{r}_i}{M} \quad (3)$$

Cartesian coordinate (x_c, y_c, z_c) of the mass center are components of the position vector $\vec{r}_c$ of the mass center.

$$x_c = \frac{\sum m_i x_i}{M}; y_c = \frac{\sum m_i y_i}{M}; z_c = \frac{\sum m_i z_i}{M} \quad (4)$$

Illustrations

Illustration 1. Center of Mass of Two Particle System

- Find expression of position vector of mass center of a system of two particles of masses m_1 and m_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$.
- Express Cartesian coordinates of mass center, if particle m_1 at point (x_1, y_1) and particle m_2 at point (x_2, y_2) .
- If you assume origin of your coordinate system at the mass center, what you conclude regarding location of the mass center relative to particles.
- Now find location of mass center of a system of two particles masses m_1 and m_2 separated by distance r .

Solution.

- Consider two particles of masses m_1 and m_2 located at position vectors $\vec{r}_1$ and $\vec{r}_2$. Let their mass center C at position vector $\vec{r}_c$.

From eq., we have

$$\vec{r}_c = \frac{\sum m_i \vec{r}_i}{M} \rightarrow \vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

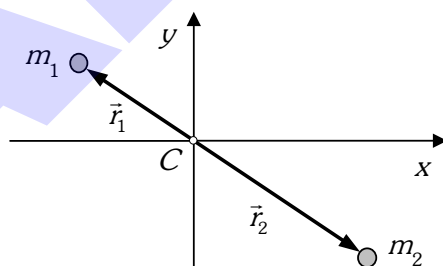
- From result obtained in part (a), we have

$$x_c = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \text{ and } y_c = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

- If we assume origin at the mass center vector $\vec{r}_c$ vanishes and we have

$$m_1 \vec{r}_1 + m_2 \vec{r}_2 = \vec{0}$$

Since either of the masses m_1 and m_2 cannot be negative, to satisfy the above equation, vectors $\vec{r}_1$ and $\vec{r}_2$ must have opposite signs. It is geometrically possible only when mass center C lies between the two particles on the line joining them as shown in the figure.

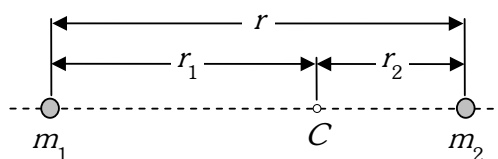


If we substitute magnitudes r_1 and r_2 of vectors $\vec{r}_1$ and $\vec{r}_2$ in the above equation, we have $m_1 r_1 = m_2 r_2$, which suggest

$$\frac{r_1}{r_2} = \frac{m_2}{m_1}$$

Now we conclude that mass center of two particle system lies between the two particles on the line joining them and divide the distance between them in inverse ratio of masses of the particles.

- (d) Consider two particles masses m_1 and m_2 at distance r from each other. Their mass center C must lie in between them on the line joining them. Let distances of these particles from the mass center are r_1 and r_2 .



Since mass center of two particle system lies between the two particles on the line joining them and divide the distance between them in inverse ratio of masses of the particles, we can write

$$r_1 = \frac{m_2 r}{m_1 + m_2} \text{ and } r_2 = \frac{m_1 r}{m_1 + m_2}$$

Illustration 2. Mass centre of several particles

Find position vectors of mass center of a system of three particles of masses 1 kg, 2 kg and 3 kg located at position vectors $\vec{r}_1 = (4\hat{i} + 2\hat{j} - 3\hat{k})\text{m}$, $\vec{r}_2 = (\hat{i} - 4\hat{j} + 2\hat{k})\text{m}$ and $\vec{r}_3 = (2\hat{i} - 2\hat{j} + \hat{k})\text{m}$ respectively.

Solution.

From eq., we have

$$\vec{r}_c = \frac{\sum m_i \vec{r}_i}{M} \rightarrow \vec{r}_c = \frac{1(4\hat{i} + 2\hat{j} - 3\hat{k}) + 2(\hat{i} - 4\hat{j} + 2\hat{k}) + 3(2\hat{i} - 2\hat{j} + \hat{k})}{1+2+3} = 2\hat{i} - 2\hat{j} + \frac{2}{3}\hat{k}$$

Center of Mass of an Extended Body or Continuous Distribution of Mass

An extended body is collection of infinitely large number of particles so closely located that we neglect separation between them and assume the body as a continuous distribution of mass. A rigid body is an extended body in which relative locations of all the particles remain unchanged under all circumstances. Therefore a rigid body does not get deformed under any circumstances.

Let an extended body is shown as a continuous distribution of mass by the shaded object in the figure. Consider an infinitely small portion of mass dm of this body. It is called a mass element and is shown at position given by position vector $\vec{r}$. Total mass M of the body is $M = \int dm$. The mass center C is assumed at position given by position vector $\vec{r}_c$. Position vector of centre of mass of such a body is given by the following equation.

$$\vec{r}_c = \frac{\int \vec{r} dm}{M} \quad (5)$$

Cartesian coordinate (x_c, y_c, z_c) of the mass center are components of the position vector $\vec{r}_c$ of the mass center.

$$x_c = \frac{\int x dm}{M}; y_c = \frac{\int y dm}{M}; z_c = \frac{\int z dm}{M} \quad (6)$$

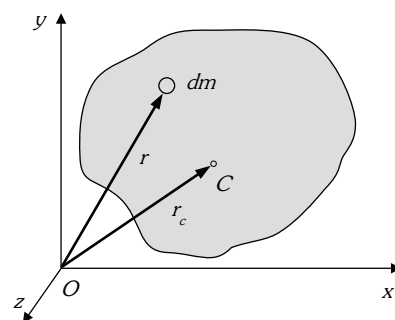
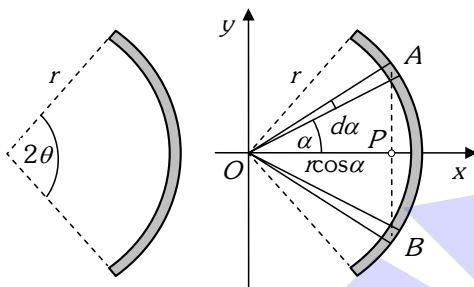


Illustration 3. Mass Center of a system of a segment of a uniform circular rod (arc)

Find location of mass center of a thin uniform rod bent into shape of an arc.

Solution. Consider a thin rod of uniform line mass density λ (mass per unit length) and radius r subtending angle 2θ on its center O . The angle bisector OP is the line of symmetry, and mass center lies on it. Therefore if we assume the angle bisector as one of the coordinate axes say x -axis, y -coordinate of mass center becomes zero.



Let two very small segments A and B located symmetric to the line of symmetry (x -axis). Mass center of these two segments is on P at a distance $x = r \cos \alpha$ from center O . Total mass of these two elements is $dm = 2\lambda r d\alpha$. Now using eq., we have

$$x_c = \frac{\int x dm}{M} \rightarrow x_c = \frac{\int_{-\theta}^{\theta} (r \cos \alpha)(2\lambda r d\alpha)}{\lambda r 2\theta} = \frac{r \sin \theta}{\theta}$$

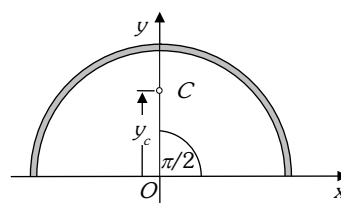
Mass center of a thin uniform arc shaped rod of radius r subtending angle 2θ at the center lies on its angle bisector at distance OC from the center.

$$OC = \frac{r \sin \theta}{\theta}$$

Illustration 4. Find coordinates of mass center of a semicircular ring of radius r placed symmetric to the y -axis of a Cartesian coordinate system.

Solution. The y -axis is the line of symmetry, therefore mass center of the ring lies on it making x coordinate zero.

Distance OC of mass center from center is given by the result obtained in Illustration 4. Making use of this result, we have

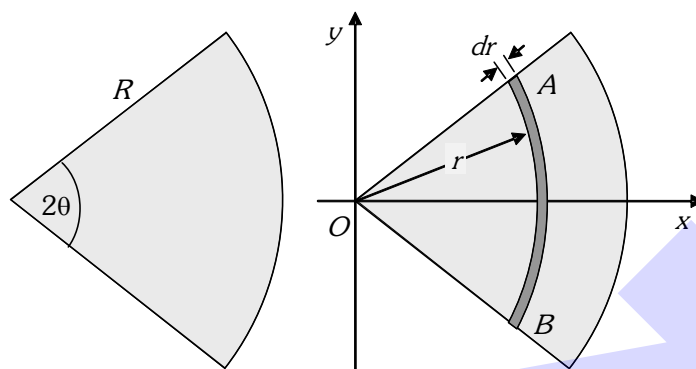


$$OC = \frac{r \sin \theta}{\theta} \rightarrow y_c = \frac{r \sin(\pi/2)}{\pi/2} = \frac{2r}{\pi}$$

Illustration 5. Mass Center of a sector of a uniform circular plate

Find location of mass center of a sector of a thin uniform plate.

Solution. Consider a sector of a thin uniform plate of surface mass density σ (mass per unit area) and radius r subtending angle 2θ on its center.



Let a thin arc of radius r and width dr be an infinitely small part of the sector. Mass dm of the arc AB equals to product of mass per unit area and area of the arc.

$$dm = \sigma(2r\theta dr) = 2\sigma r\theta dr$$

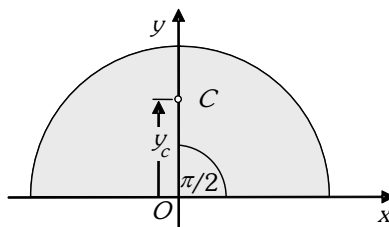
Due to symmetry mass center of this arc must be on the angle bisector i.e. on x -axis at distance $x = \frac{r \sin \theta}{\theta}$.

Using above two information in eq. , we obtain the mass center of the sector.

$$x_c = \frac{\int x dm}{M} \rightarrow x_c = \frac{\int_0^R \left(\frac{r \sin \theta}{\theta} \right) (2\sigma r \theta dr)}{\sigma (\text{Area of the sector})} = \frac{\int_0^R \left(\frac{r \sin \theta}{\theta} \right) (2\sigma r \theta dr)}{\sigma r \theta^2} = \frac{2r \sin \theta}{3\theta}$$

Illustration 6. Find coordinates of mass center of a uniform semicircular plate of radius r placed symmetric to the y -axis of a Cartesian coordinate system, with centre at origin.

Solution. The y -axis is the line of symmetry, therefore mass center of the plate lies on it making x -coordinate zero.



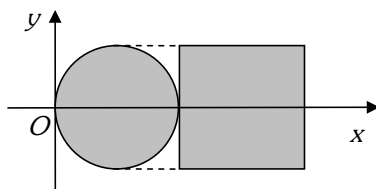
Distance OC of mass center from center is given by the result obtained in Illustration 7. Making use of this result, we have

$$OC = \frac{2r \sin \theta}{3\theta} \rightarrow y_c = \frac{2r \sin(\pi/2)}{3\pi/2} = \frac{4r}{3\pi}$$

Illustration 7. Mass Center of composite bodies

A composite body is made of joining two or more bodies. Find mass center of the following composite body made by joining a uniform disk of radius r and a uniform square plate of side r of the same mass per unit area.

Solution. To find mass center the component bodies are assumed particle of masses equal to corresponding bodies located on their respective mass centers. Then we use equation to find coordinates of the mass center of the composite body.



To find mass center of the composite body, we first have to calculate masses of the bodies, because their mass distribution is given.

If we denote surface mass density (mass per unit area) by σ , masses of the bodies are

$$\text{Mass of the disk} \quad m_d = \text{Mass per unit area} \times \text{Area} = \sigma(\pi r^2) = \sigma\pi r^2$$

$$\text{Mass of the square plate} \quad m_p = \text{Mass per unit area} \times \text{Area} = \sigma(r^2) = \sigma r^2$$

$$\text{Location of mass center of the disk} \quad x_d = \text{Center of the disk} = r \quad \text{and} \quad y_d = 0$$

$$\text{Location of mass center of the square plate} \quad x_p = \text{Center of the surface plate} = 3r \quad \text{and} \quad y_d = 0$$

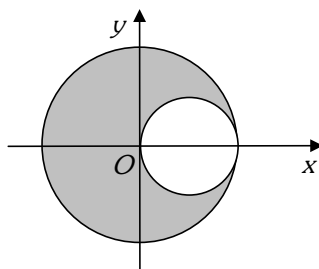
Using eq., we obtain coordinates (x_c, y_c) of the composite body.

$$x_c = \frac{m_d x_d + m_p x_p}{m_d + m_p} = \frac{r(\pi + 3)}{(\pi + 1)} \quad \text{and} \quad y_c = \frac{m_d y_d + m_p y_p}{m_d + m_p} = 0$$

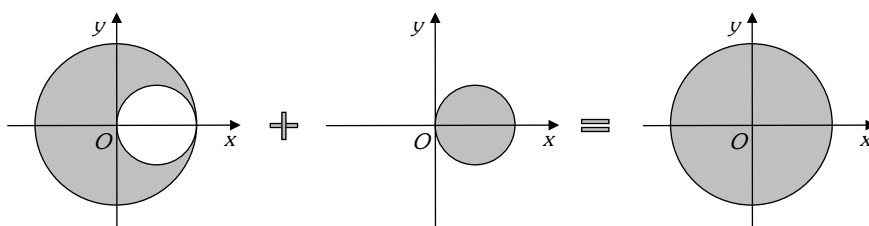
$$\text{Coordinates of the mass center are} \quad \left(\frac{r(\pi + 3)}{(\pi + 1)}, 0 \right)$$

Illustration 8. Mass Center of truncated bodies

A truncated body is made by removing a portion of a body. Find mass center of the following truncated disk made by removing disk of radius equal to half of the original disk as shown in the figure. Radius of the original uniform disk is r .



Solution. To find mass center of truncated bodies we can make use of superposition principle that is, if we add the removed portion in the same place we obtain the original body. The idea is illustrated in the following figure.



The removed portion is added to the truncated body keeping their location unchanged relative to the coordinate frame.

Denoting masses of the truncated body, removed portion and original body by m_{tb} , m_{rp} and m_{ob} and location of their mass centers by x_{tb} , x_{rp} and x_{ob} , we can write

$$m_{tb}x_{tb} + m_{rp}x_{rp} = m_{ob}x_{ob}$$

From the above equation we obtain position co-ordinate x_{tb} of the mass center of the truncated body.

$$x_{tb} = \frac{m_{ob}x_{ob} - m_{rp}x_{rp}}{m_{tb}} \quad (1)$$

Denoting mass per unit area by σ , we can express the masses m_{tb} , m_{rp} and m_{ob} .

Mass of truncated body $m_{tb} = \sigma \left\{ \pi \left(r^2 - \frac{r^2}{4} \right) \right\} = \frac{3\sigma\pi r^2}{4}$

Mass of the removed portion $m_{rp} = \frac{\sigma\pi r^2}{4}$

Mass of the original body $m_{ob} = \sigma\pi r^2$

Mass center of the truncated body x_{tb}

Mass center of the removed portion $x_{rp} = \frac{r}{2}$

Mass center of the original body $x_{ob} = 0$

Substituting the above values in equation (1), we obtain the mass the center of the truncated body.

$$x_{tb} = \frac{m_{ob}x_{ob} - m_{rp}x_{rp}}{m_{tb}} = \frac{(\sigma\pi r^2) \times 0 - \left(\frac{\sigma\pi r^2}{4} \right) \left(\frac{r}{2} \right)}{\frac{3\sigma\pi r^2}{4}} = -\frac{r}{6}$$

Mass center of the truncated body is at point $\left(-\frac{r}{6}, 0 \right)$

1.2 Center of Mass Frame of Reference or Centroidal Frame

Center of mass frame of reference or centroidal frame is reference frame assume attached with the mass center of the system at its origin. It moves together with the mass center.

It is a special frame and presents simple interpretations and solutions to several phenomena. Let us first discuss some of its fundamental properties.

In centroidal frame center of mass is assumed at the origin, therefore position vector, velocity and acceleration of the mass center in centroidal frame all become zero.

- Sum of mass moments in centroidal frame vanishes.

Mass moment of a particle is product of mass of the particle and its position vector.

$$\sum m_i \vec{r}_{i/c} = \vec{0} \quad \text{or} \quad m_1 \vec{r}_{1/c} + m_2 \vec{r}_{2/c} + \dots + m_i \vec{r}_{i/c} + \dots + m_n \vec{r}_{n/c} = \vec{0} \quad (7)$$

- Total linear momentum of the system in centroidal frame vanishes..

$$\sum m_i \vec{v}_{i/c} = \vec{0} \quad \text{or} \quad m_1 \vec{v}_{1/c} + m_2 \vec{v}_{2/c} + \dots + m_i \vec{v}_{i/c} + \dots + m_n \vec{v}_{n/c} = \vec{0} \quad (8)$$

Illustrations

Illustration 9. Motion of Mass Center in One Dimension

A jeep of mass 2400 kg is moving along a straight stretch of road at 80 km/h. It is followed by a car of mass 1600 kg moving at 60 km/h.

- How fast is the center of mass of the two cars moving?
- Find velocities of both the vehicles in centroidal frame.

Solution.

$$(a) \text{ Velocity of the mass center } \vec{v}_c = \frac{m_{jeep} \vec{v}_{jeep} + m_{car} \vec{v}_{car}}{m_{jeep} + m_{car}}$$

Assuming direction of motion in the positive x-direction, we have

$$\vec{v}_c = \frac{m_{jeep} \vec{v}_{jeep} + m_{car} \vec{v}_{car}}{m_{jeep} + m_{car}} \rightarrow \vec{v}_c = \frac{2400 \times 80 + 1600 \times 60}{2400 + 1600} = 72 \text{ km/h}$$

- Velocity of the jeep in centroidal frame

$$v_{jeep/c} = 80 - 72 = 8 \text{ km/h in positive x-direction.}$$

$$\text{Velocity of the car in centroidal frame } v_{car/c} = 60 - 72 = -12 \text{ km/h}$$

12 km/h negative x-direction.

Illustration 10. Motion of Mass Center in Vector Form

A 2.0 kg particle has a velocity of $\vec{v}_1 = (2.0\hat{i} - 3.0\hat{j})$ m/s, and a 3.0 kg particle has a velocity $\vec{v}_2 = (1.0\hat{i} + 6.0\hat{j})$ m/s.

- How fast is the center of mass of the particle system moving?
- Find velocities of both the particles in centroidal frame.

Solution. (a) Velocity of the mass center $\vec{v}_c = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$

$$\vec{v}_c = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \rightarrow \vec{v}_c = \frac{2(2.0\hat{i} - 3.0\hat{j}) + 3(1.0\hat{i} + 6.0\hat{j})}{2 + 3} = (1.4\hat{i} + 2.4\hat{j}) \text{ m/s}$$

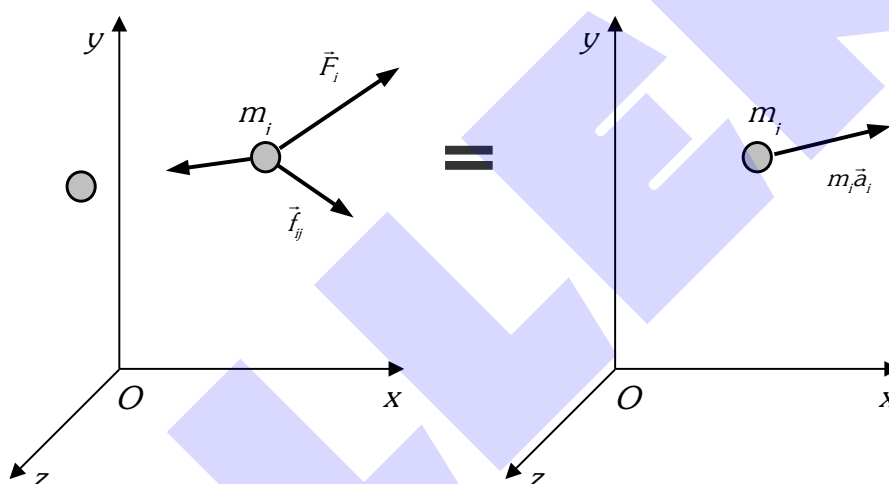
(b) Velocity of the first particle in centroidal frame

$$\vec{v}_{1/c} = \vec{v}_1 - \vec{v}_c \rightarrow \vec{v}_{1/c} = (2.0\hat{i} - 3.0\hat{j}) - (1.4\hat{i} + 2.4\hat{j}) = 0.6\hat{i} - 5.4\hat{j} \text{ m/s}$$

Velocity of the second particle in centroidal frame

$$\vec{v}_{2/c} = \vec{v}_2 - \vec{v}_c \rightarrow \vec{v}_{2/c} = (1.0\hat{i} + 6.0\hat{j}) - (1.4\hat{i} + 2.4\hat{j}) = -(0.4\hat{i} + 3.6\hat{j}) \text{ m/s}$$

1.3 Application of Newton's Laws of Motion to a System of Particles



In order to write equation of motion for a system of particles, we begin by applying Newton's second law to an individual particle.

Consider i^{th} particle of mass m_i . Internal force applied on it by j^{th} particle is shown by $\vec{f}_{ij}$. Other particles of the system may also apply internal forces on it. One of them is shown in the figure by an unlabeled vector. In addition to these internal forces, external forces may also be applied on it by bodies out side the system. Resultant of all these external forces is shown by vector $\vec{F}_i$. If under the action of these forces this particles has acceleration $\vec{a}_i$ relative to an inertial frame Oxyz, its free body diagram and kinetic diagram can be represented by the following figure and Newton's second law can be written by the following equation.

$$\vec{F}_i + \sum \vec{f}_{ij} = m_i \vec{a}_i$$

In similar fashion, we can write Newton's second law for all the particles of the system. These equations are

For 1st particle $\vec{F}_1 + \sum \vec{f}_{1j} = m_1 \vec{a}_1$

For 2nd particle $\vec{F}_2 + \sum \vec{f}_{2j} = m_2 \vec{a}_2$

.....

For ith particle $\vec{F}_i + \sum \vec{f}_{ij} = m_i \vec{a}_i$

.....

For nth particle $\vec{F}_n + \sum \vec{f}_{nj} = m_n \vec{a}_n$

Every internal force $\vec{f}_{ij}$ on particle m_i due to particle m_j and $\vec{f}_{ji}$ on the particle m_j due to particle m_i constituting Newton's third law pair must be equal in magnitude and opposite in direction, therefore the sum all these internal forces for all the particles must be zero. Keeping this fact in mind and denoting the mass of the whole system by M and acceleration of the mass center C by $\vec{a}_C$ relative to the inertial frame, Newton's second law representing translation motion of the system of particles can be represented by the following equation.

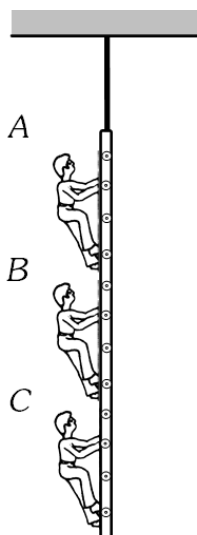
$$\sum \vec{F}_i = \sum (m_i \vec{a}_i) = M \vec{a}_C$$

$$\sum \vec{F}_i = M \vec{a}_C = \frac{d\vec{p}_C}{dt}$$

Illustrations

Illustration 11. Newton's Laws of Motion and System of Particles

A ladder of mass 20 kg is hanging from ceiling as shown in figure. Three men A, B and C of masses 40 kg, 60 kg, and 50 kg are climbing the ladder. Man A is climbing with upward retardation 2 m/s^2 , B is climbing up with a constant speed of 0.5 m/s and C is climbing with upward acceleration of 1 m/s^2 . Find the tension in the string supporting the ladder.



Solution. External forces acting on the system are weights of the men, weight of the ladder and tension supporting the ladder. Denoting masses of men A, B, C and ladder by m_A , m_B , m_C and m_L , acceleration due to gravity by g , tension in the string by T and accelerations of the men A, B, C and ladder by a_A , a_B , a_C and a_L respectively, we can write the following equation according to equation.

$$\Sigma \vec{F}_i = \Sigma (m_i \vec{a}_i) \rightarrow T - m_A g - m_B g - m_C g - m_L g = m_A a_A + m_B a_B + m_C a_C + m_L a_L$$

Substituting given values of masses $m_A = 40 \text{ kg}$, $m_B = 60 \text{ kg}$, $m_C = 50 \text{ kg}$, $m_L = 20 \text{ kg}$

given values of accelerations $g = 10 \text{ m/s}^2$, $a_A = -2 \text{ m/s}^2$, $a_B = 0 \text{ m/s}^2$, $a_C = 1 \text{ m/s}^2$, and $a_L = 0 \text{ m/s}^2$

we obtain $T - 400 - 600 - 500 - 200 = -80 + 0 + 50 + 0$

$$T = 1670 \text{ N}$$

2.0 IMPULSE OF A FORCE

Net force applied on a rigid body changes momentum i.e. amount of motion of that body. A net force for a longer duration cause more change in momentum than the same force acting for shorter duration. Therefore duration in which a force acts on a body together with magnitude and direction of the force decide effect of the force on the change in momentum of the body.

Linear impulse or simply impulse of a force is defined as integral of the force with respect to time.

If a force $\vec{F}$ acts on a body, its impulse in a time interval from t_i to t_f is given by the following equation.

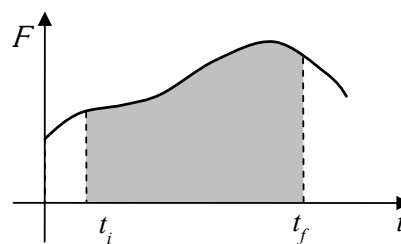
$$\vec{I}_{mp} = \int_{t_i}^{t_f} \vec{F} dt$$

If the force is constant, its impulse equals to product of the force vector $\vec{F}$ and time interval Δt .

$$\vec{I}_{mp} = \vec{F}(\Delta t)$$

For one-dimensional force, impulse equals to area between force-time

graph and the time axis. In the given figure is shown how a force F along x-axis varies with time t . Impulse of this force in time interval t_i to t_f equals to area of the shaded portion.



If several forces $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots, \vec{F}_n$ act on a body in a time interval, the total impulse $\vec{I}_{mp}$ of all these forces equals to impulse of the net force.

$$\vec{I}_{mp} = \int_{t_i}^{t_f} \vec{F}_1 dt + \int_{t_i}^{t_f} \vec{F}_2 dt + \dots + \int_{t_i}^{t_f} \vec{F}_n dt = \int_{t_i}^{t_f} (\vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n) dt$$

Impulse is measured in newton-second.

Dimensions of impulse are MLT^{-1}

Illustrations

Illustration 12. Calculate impulse of force $\vec{F} = (3t^2\hat{i} - (2t-1)\hat{j} + 2\hat{k})\text{N}$ over the time interval from $t = 1\text{ s}$ to $t = 3\text{ s}$.

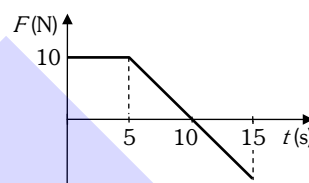
Solution.

$$\vec{I}_{\text{mp}} = \int_{t_i}^{t_f} \vec{F} dt \rightarrow \vec{I}_{\text{mp}} = \int_1^3 (3t^2\hat{i} - (2t-1)\hat{j} + 2\hat{k}) dt$$

$$= \left[t^3\hat{i} - (t^2 - t)\hat{j} + 2t\hat{k} \right]_1^3 = \hat{i}[t^3]_1^3 - \hat{j}[(t^2 - t)]_1^3 + \hat{k}[2t]_1^3 \text{ N-s}$$

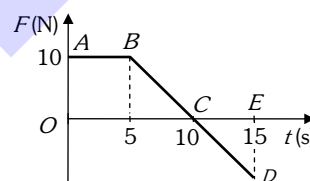
Illustration 13. A one-dimensional force F varies with time according to the given graph. Calculate impulse of the force in following time intervals.

- From $t = 0\text{ s}$ to $t = 10\text{ s}$.
- From $t = 10\text{ s}$ to $t = 15\text{ s}$.
- From $t = 0\text{ s}$ to $t = 15\text{ s}$.



Solution. For one-dimensional force, impulse equals to area between force-time graph and the time axis.

- $I_{0 \rightarrow 10} = \text{Area of trapezium OABC} = 75\text{ N-s}$
- $I_{10 \rightarrow 15} = -\text{Area of triangle CDE} = -25\text{ N-s}$
- $I_{0 \rightarrow 15} = \text{Area of trapezium OABC} - \text{Area of triangle CDE} = 50\text{ N-s}$



2.1 Impulse Momentum Principle

Consider body of mass m in translational motion. When it is moving with velocity $\vec{v}$, net external force acting on it is $\vec{F}$. Equation of motion as suggested by Newton's second law can be written in the form

$$\vec{F} dt = d(m\vec{v})$$

If the force acts during time interval from t_i to t_f and velocity of the body changes from $\vec{v}_i$ to $\vec{v}_f$, integrating the above equation with time over the interval from t_i to t_f we have

$$\int_{t_i}^{t_f} \vec{F} dt = m\vec{v}_f - m\vec{v}_i$$

Here left hand side of the above equation is impulse $\vec{I}_{\text{mp}}$ of the net force $\vec{F}$ in time interval from t_i to t_f , and quantities $m\vec{v}_i$ and $m\vec{v}_f$ on the right hand side are linear momenta of the particle at instants t_i and t_f . If we denote them by symbols $\vec{p}_i$ and $\vec{p}_f$, the above equation can be written as

$$\vec{I}_{\text{mp}} = \vec{p}_f - \vec{p}_i$$

The idea expressed by the above equation is known as impulse momentum principle. It states that change in the momentum of a body in a time interval equals to the impulse of the net force acting on the body during the concerned time interval.

For the ease of application to physical situations the above equation is rearranged as

$$\vec{p}_i + \vec{I}_{mp} = \vec{p}_f$$

This equation states that impulse of a force during a time interval when added to momentum of a body at the beginning of an interval of time we get momentum of the body at the end of the interval concerned.

Since impulse and momentum both are vector quantities, the impulse momentum theorem can be expressed by there scalar equations making use of Cartesian components.

$$p_{1x} + \sum I_{mp,x} = p_{2x} \quad p_{1y} + \sum I_{mp,y} = p_{2y} \quad p_{1z} + \sum I_{mp,z} = p_{2z}$$

The impulse momentum principle is deduced here for a single body moving relative to an inertial frame, therefore impulses of only physical forces are considered. If we are using a non-inertial reference frame, impulse of corresponding pseudo force must also be considered in addition to impulse of the physical forces.

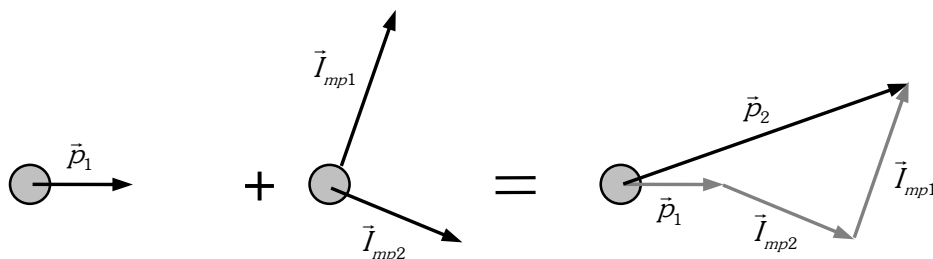
How to apply Impulse Momentum Principle

The impulse momentum principle is deduced here for a single body, therefore it is recommended at present to use it for a single body. To use this principle the following steps should be followed.

- (i) Identify the initial and final positions as position 1 and 2 and show momenta $\vec{p}_1$ and $\vec{p}_2$ of the body at these instants.
- (ii) Show impulse of each force acting on the body at an instant between positions 1 and 2.
- (iii) Use the impulse obtained in step (ii) and momenta obtained in step (i) into equation

$$\vec{p}_i + \vec{I}_{mp} = \vec{p}_f$$

Consider a particle moving with momentum $\vec{p}_1$ in beginning. It is acted upon by two forces, whose impulses in a time interval are $\vec{I}_{mp1}$ and $\vec{I}_{mp2}$. As a result, at the end of the time interval, momentum of the particle becomes $\vec{p}_2$. This physical situation is shown in the following diagram. Such a diagram is known as impulse momentum diagram.



Illustrations

Illustration 14. A particle of mass 2 kg is moving with velocity $\vec{v}_o = (2\hat{i} - 3\hat{j})$ m/s in free space. Find its velocity 3 s after a constant force $\vec{F} = (3\hat{i} + 4\hat{j})$ N starts acting on it.

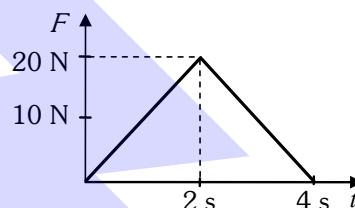
Solution. $\vec{p}_f = \vec{p}_i + \vec{I}_{mp} \rightarrow m\vec{v}_f = m\vec{v}_o + \vec{F}\Delta t$

Substituting given values, we have

$$2\vec{v}_f = 2(2\hat{i} - 3\hat{j}) + (3\hat{i} + 4\hat{j}) \times 3 = 13\hat{i} + 6\hat{j}$$

$$\vec{v}_f = (6.5\hat{i} + 3\hat{j}) \text{ m/s}$$

Illustration 15. A box of mass $m = 2$ kg resting on a frictionless horizontal ground is acted upon by a horizontal force F , which varies as shown. Find speed of the particle when the force ceases to act.

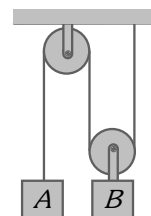


Solution. $\vec{p}_f = \vec{p}_i + \vec{I}_{mp} \rightarrow m\vec{v}_f = m\vec{v}_i + \int_{t_i}^{t_f} \vec{F} dt$

$$2v = 2 \times 0 + \frac{1}{2} \times 20 \times 4$$

$$v = 20 \text{ m/s}$$

Illustration 16. Two boxes A and B of masses m and M interconnected by an ideal rope and ideal pulleys are held at rest as shown. When it is released, box B accelerates downwards. Find velocities of box A and B as function of time t after system has been released.

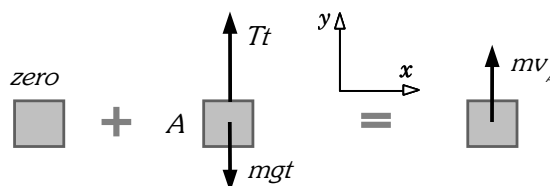
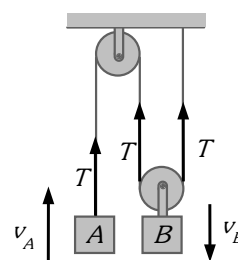


Solution. We first explore relation between accelerations a_A and a_B of the boxes A and B, which can be written either by using constrained relation or method of virtual work or by inspection.

$$v_A = 2v_B \dots (i)$$

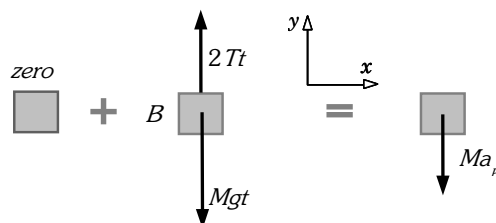
Applying impulse momentum principle to box A

$$p_{2y} = p_{1y} + \sum I_{mp,y} \rightarrow Mv_A = 0 + Tt - mgt \dots (ii)$$



Applying impulse momentum principle to box B

$$p_{2y} = p_{1y} + \sum I_{mp,y} \rightarrow mv_B = 0 + Mgt - 2Tt \quad \dots(iii)$$

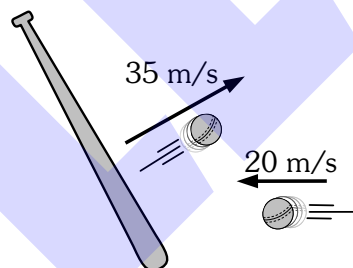


From equations (i), (ii) and (iii), we have and

Illustrations

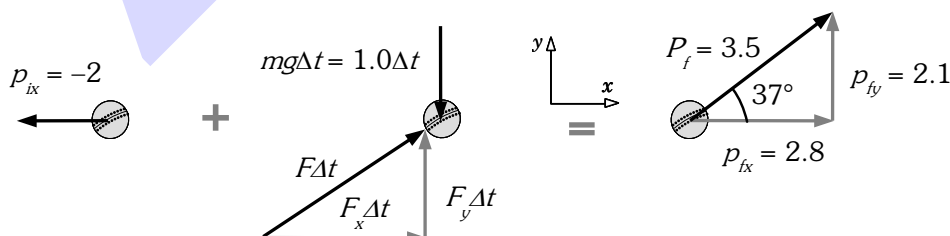
Illustration 17. A 100 gm ball moving horizontally with 20 m/s is struck by a bat, as a result it starts moving with a speed of 35 m/s at an angle of 37° above the horizontal in the same vertical plane as shown in the figure.

- Find the average force exerted by the bat if duration of impact is 0.30 s.
- Find the average force exerted by the bat if duration of impact is 0.03 s.
- Find the average force exerted by the bat if duration of impact is 0.003 s.
- What do you conclude for impulse of weight of the ball as duration of contact decreases?



Solution.

The impulse momentum diagram of the ball is shown in the figure below. Here F , mg , and Δt represent the average value of the force exerted by the bat, weight of the ball and the time interval.



Applying principle of impulse and momentum in x-direction, we have

$$p_{fx} = p_{ix} + \sum I_{mp,x} \rightarrow -2 + F_x\Delta t = 2.8$$

$$F_x = \frac{4.8}{\Delta t} \text{ N} \quad \dots(i)$$

Applying principle of impulse and momentum in y-direction, we have

$$p_{iy} = p_{iy} + \sum I_{mp,y} \rightarrow 0.0 + F_y \Delta t - 1.0 \Delta t = 2.1$$

$$F_y = \left(\frac{2.1}{\Delta t} + 1.0 \right) \text{ N} \quad \dots(\text{ii})$$

(a) Substituting $\Delta t = 0.30$ s, in equations (i) and (ii), we find $\vec{F} = 16\hat{i} + 8\hat{j}$ N

(b) Substituting $\Delta t = 0.03$ s, in equations (i) and (ii), we find $\vec{F} = 160\hat{i} + 71\hat{j}$ N

(c) Substituting $\Delta t = 0.003$ s, in equations 1 and 2, we find $\vec{F} = 1600\hat{i} + 701\hat{j}$ N

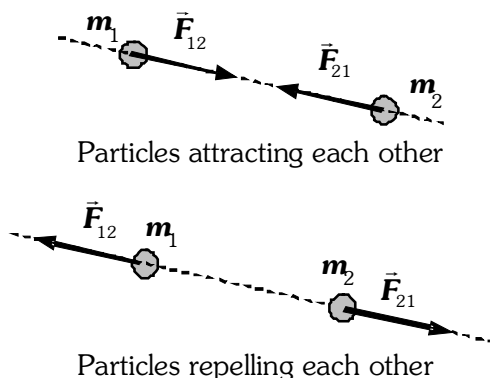
(d) It is clear from the above results that as the duration of contact between the ball and the bat decreases, effect of the weight of the ball also decreases as compared with that of the force of the bat and for sufficiently short time interval, it can be neglected.

3.0 Principle of Conservation of linear momentum

The principle of conservation of linear momentum or simply conservation of momentum for two or more interacting bodies is one of guiding principles of the classical as well as the modern physics.

To understand this principle, we first discuss a system of two interacting bodies, and then extend the ideas developed to a system consisting of many interacting bodies.

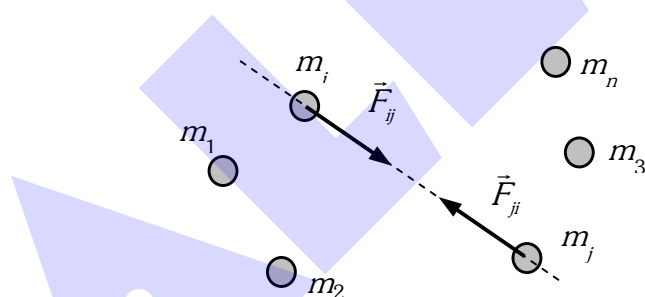
Consider a system of two particles of mass m_1 and m_2 . Particle m_1 attracts m_2 with a force $\vec{F}_{12}$ and m_2 attracts m_1 with a force $\vec{F}_{21}$. These forces have equal magnitudes and opposite directions as shown in the figure. If the bodies are let free i.e. without any external force acting on any of them, each of them move and gain momentum equal to the impulse of the force of interaction. Since equal and opposite interaction forces act on both of them for the same time interval, the momenta gained by them are equal in magnitude and opposite in direction resulting no change in total momentum of the system.



However, if an external force acts on any one of them or different forces with a nonzero resultant act on both of them, the total momentum of the bodies will certainly change. If the system undergoes an impulsive motion, total momentum will change only under the action of external impulsive force or forces. Internal impulsive forces also exist in pairs of equal and opposite forces and cannot change the total momentum of the system. Non-impulsive forces if act cannot change momentum of the system by appreciable amount. For example, gravity is a non-impulsive force, therefore in the process of collision between two bodies near the earth the total momentum remains conserved.

The total momentum of a system of two interacting bodies remains unchanged under the action of the forces of interaction between them. It can change only if a net impulse of external force is applied.

In similar way we may conceive a model of a system of n interacting particles having masses $m_1, m_2, \dots, m_i, \dots, m_j, \dots$ and m_n respectively. The forces of interaction $\vec{F}_{ij}$ and $\vec{F}_{ji}$ between m_i and m_j are shown in the figure. Since internal forces exist in pairs of equal and opposite forces, in any time interval of concern each of them have a finite impulse but their total impulse is zero. Thus if the system is let free, in any time interval momentum of every individual particle changes but the total momentum of the system remains constant.



System of n interacting particles.

It can change only if external forces are applied to some or all the particles. Under the action of external forces, the change in total momentum of the system will be equal to the net impulse of all the external forces.

Thus, total momentum of a system of particles cannot change under the action of internal forces and if net impulse of the external forces in a time interval is zero, the total momentum of the system in that time interval will remain conserved.

$$\sum \vec{p}_{initial} = \sum \vec{p}_{final}$$

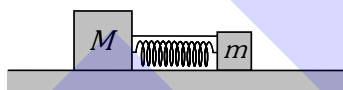
The above statement is known as the principle of conservation of momentum. It is applicable only when the net impulse of all the external forces acting on a system of particles becomes zero in a finite time interval. It happens in the following conditions.

- When no external force acts on any of the particles or bodies.
- When resultant of all the external forces acting on all the particles or bodies is zero.
- In impulsive motion, where time interval is negligibly small, the direction in which no impulsive forces act, total component of momentum in that direction remains conserved.

Since force, impulse and momentum are vectors, component of momentum of a system in a particular direction is conserved, if net impulse of all external forces in that direction vanishes.

Illustrations

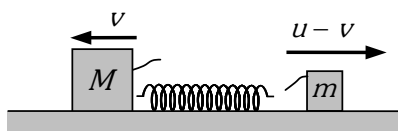
Illustration 18. Two blocks of masses m and M are held against a compressed spring on a frictionless horizontal floor with the help of a light thread. When the thread is cut, the smaller block leaves the spring with a velocity u relative to the larger block. Find the recoil velocity of the larger block.



Solution.

When the thread is cut, the spring pushes both the block, and impart them momentum. The forces applied by the spring on both the block are internal forces of the two-block system. External forces acting on the system are weights and normal reactions on the blocks from the floor. These external forces have zero net resultant of the system. In addition to this fact no external force acts on the system in horizontal direction, therefore, horizontal component of the total momentum of the system remains conserved.

Velocities of both the objects relative to the ground (inertial frame) are shown in the adjoining figure.

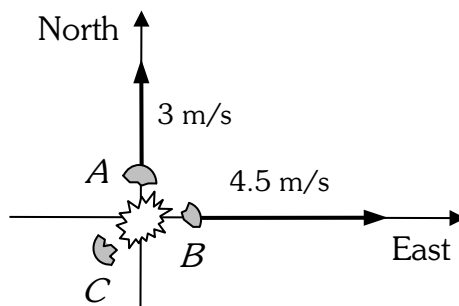


Since before the thread is cut system was at rest, its total momentum was zero. Principle of conservation of momentum for the horizontal direction yields

$$\sum_{i=1}^n p_{\text{horizontal}} = 0 \rightarrow -Mv + m(u - v) = 0$$

$$v = \frac{mu}{M + m}$$

Illustration 19. A shell fired vertically up, when reaches its highest point, explodes into three fragments A, B and C of masses $m_A = 4$ kg, $m_B = 2$ kg and $m_C = 3$ kg. Immediately after the explosion, A is observed moving with velocity $v_A = 3$ m/s towards north and B with a velocity $v_B = 4.5$ m/s towards east as shown in the figure. Find the velocity v_C of the piece C.



Solution.

Explosion takes negligible duration; therefore, impulse of gravity, which is a finite external force, can be neglected. The pieces fly off acquiring above-mentioned velocities due to internal forces developed due to expanding gases produced during the explosion. The forces applied by the expanding gases are internal forces; hence, momentum of the system of the three pieces remains conserved during the explosion and total momentum before and after the explosion are equal.

Assuming the east as positive x-direction and the north as positive y-direction, the momentum vectors $\vec{p}_A$ and $\vec{p}_B$ of pieces A and B become

$$\vec{p}_A = m_A v_A \hat{j} = 12 \text{ kg-m/s and } \vec{p}_B = m_B v_B \hat{i} = 9 \text{ kg-m/s}$$

Before the explosion, momentum of the shell was zero, therefore from the principle of conservation of momentum, the total momentum of the fragments also remains zero.

$$\vec{p}_A + \vec{p}_B + \vec{p}_C = \vec{0} \rightarrow \vec{p}_C = -(9\hat{i} + 12\hat{j})$$

From the above equation, velocity of the piece C is

$$\vec{v}_C = \frac{\vec{p}_C}{m_C} = -(3\hat{i} + 4\hat{j}) = 5 \text{ m/s, } 53^\circ \text{ south of west.}$$

Illustration 20. In free space, three identical particles moving with velocities $v_o \hat{i}$, $-3v_o \hat{j}$ and $5v_o \hat{k}$ collide successively with each other to form a single particle. Find velocity vector of the particle formed.

Solution.

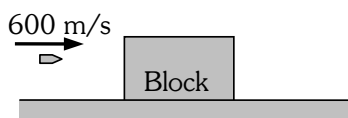
Let m be the mass of a single particle before any of the collisions. The mass of particle formed after collisions must be $3m$. In free space, no external forces act on any of the particles, their total momentum remains conserved.

Applying principle of conservation of momentum, we have

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}} \rightarrow mv_o \hat{i} - 3mv_o \hat{j} + 5mv_o \hat{k} = 3m\vec{v}$$

$$\vec{v} = \frac{1}{3}v_o (\hat{i} - 3\hat{j} + 5\hat{k}) \text{ m/s}$$

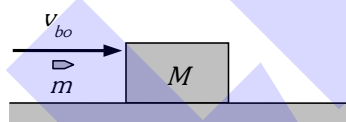
Illustration 21. A bullet of mass 50 g moving with velocity 600 m/s hits a block of mass 1.0 kg placed on a rough horizontal ground and comes out of the block with a velocity of 400 m/s. The coefficient of friction between the block and the ground is 0.25. Neglect loss of mass of the block as the bullet pierces through it.



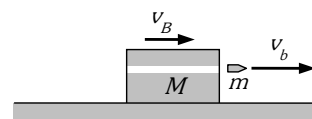
- In spite of the fact that friction acts as an external force, can you apply principle of conservation of momentum during interaction of the bullet with the block?
- Find velocity of the block immediately after the bullet pierces through it.
- Find the distance the block will travel before it stops.

Solution.

- There is no net external force in the vertical direction and in the horizontal direction, only external force friction is non-impulsive, therefore momentum of the bullet-block system during their interaction remains conserved.
- Let us denote velocities of the bullet before it hits the block and immediately after it pierces through the block by v_{bo} and v_b , velocity of the block immediately after the bullet pierces through it is v_B and masses of the bullet and the block by m and M respectively. These are shown in the adjacent figure.



Immediately before the bullet hits the block



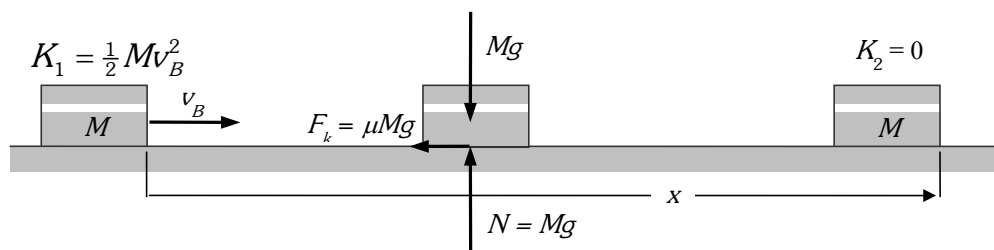
Immediately after the bullet pierces the block

Applying principle of conservation of momentum for horizontal component, we have

$$mv_{bo} = mv_b + Mv_B \rightarrow v_B = \frac{m(v_{bo} - v_b)}{M}$$

Substituting the given values, we have $v_B = 10$ m/s

- To calculate distance traveled by the block before it stops, work kinetic energy theorem has to be applied.



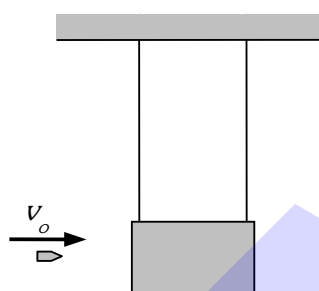
During sliding of the box on the ground only the force of kinetic friction does work.

$$W_{1 \rightarrow 2} = K_2 - K_1 \rightarrow -\mu Mgx = 0 - \frac{1}{2} Mv_B^2$$

$$x = \frac{v_B^2}{2\mu g}$$

Substituting given values, we have $x = 20$ m

Illustration 22. Ballistic Pendulum : A ballistic pendulum is used to measure speed of bullets. It consists of a wooden block suspended from fixed support.

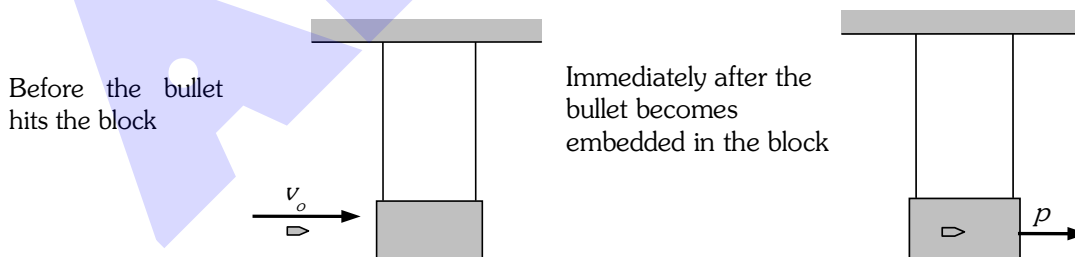


A wooden block of mass M is suspended with the help of two threads to prevent rotation while swinging. A bullet of mass m moving horizontally with velocity v_o hits the block and becomes embedded in the block.

Receiving momentum from the bullet, the bullet-block system swings to a height h . Find expression for speed of the bullet in terms of given quantities.

Solution.

When the bullet hits the block, in a negligible time interval, it becomes embedded in the block and the bullet-block system starts moving with horizontally. During this process, net force acting on the bullet-block system in vertical direction is zero and no force acts in the horizontal direction. Therefore, momentum of the bullet-block system remains conserved.



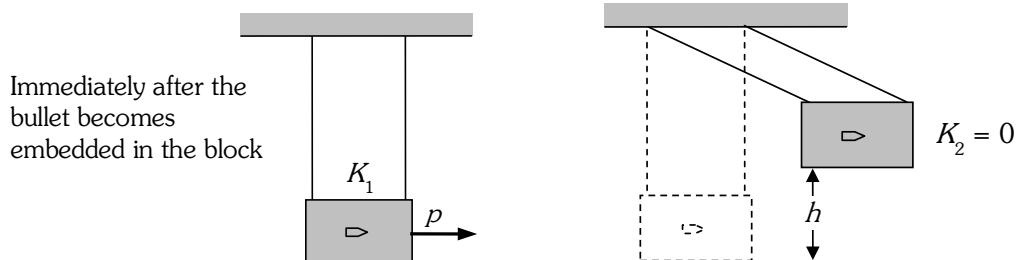
Let us denote momentum of the bullet-block system immediately after the bullet becomes embedded in the block by p and apply principle of conservation of momentum to the system for horizontal component of momentum.

$$p = mv_o$$

Using equation $K = p^2 / 2m$, we can find kinetic energy K_1 of the bullet-block system immediately after the bullet becomes embedded in the block.

$$K_1 = \frac{(mv_o)^2}{2(M+m)}$$

During swing, only gravity does work on the bullet-block system. Applying work-kinetic energy theorem during swing of the bullet-block system, we have



$$0W_{1 \rightarrow 2} = K_2 - K_1 \rightarrow -(M+m)gh = 0 - \frac{(mv_o)^2}{2(M+m)}$$

$$\text{Rearranging terms, we have } v_o = \frac{(M+m)}{m} \sqrt{2gh}$$

3.3 Application of Methods of Impulse and Momentum to a System of Particles

In a phenomenon, when a system changes its configuration, some or all of its particles change their respective locations and momenta. Sum of linear momenta of all the particles equals to the linear momentum due to translation of mass center. Principle of impulse and moment suggests net impulse of all the external forces equals to change in momentum of mass center.

$$\sum \int \vec{F}_i dt = \vec{p}_{cf} - \vec{p}_{ci}$$

Conservation of Linear momentum

The above event suggests that total linear momentum of a system of particle remains conserved in a time interval in which impulse of external forces is zero.

Total momentum of a system of particles cannot change under the action of internal forces and if net impulse of the external forces in a time interval is zero, the total momentum of the system in that time interval will remain conserved.

$$\sum \vec{p}_{initial} = \sum \vec{p}_{final} \text{ or } \vec{p}_{ci} = \vec{p}_{cf}$$

The above statement is known as the principle of conservation of momentum.

Since force, impulse and momentum are vectors, component of momentum of a system in a particular direction is conserved, if net impulse of all external forces in that direction vanishes.

During an event the net impulse of external forces in a direction is zero in the following cases.

- When no external force acts in a particular direction on any of the particles or bodies.
- When resultant of all the external forces acting in a particular direction on all the particles or bodies is zero.
- In impulsive motion, where time interval is negligibly small, the direction in which no impulsive forces act.

Illustrations

Illustration 23. No external force: Stationary mass relative to an inertial frame remains at rest

A man of mass m is standing at one end of a plank of mass M . The length of the plank is L and it rests on a frictionless horizontal ground. The man walks to the other end of the plank. Find displacement of the plank and man relative to the ground.



Solution.

Denoting x -coordinates of the man, mass center of plank and mass center of the man-plank system by x_m , x_p and x_c , we can write the following equation.

$$(\sum m_i) \vec{r}_c = \sum m_i \vec{r}_i \rightarrow (m + M) \vec{x}_c = m \vec{x}_m + M \vec{x}_p$$

Net force on the system relative to the ground is zero. Therefore mass center of the system which is at rest before the man starts walking, remains at rest ($\Delta \vec{x}_c = 0$) after while the man walks on the plank.

$$(\Delta \vec{x}_c = 0) \rightarrow m \Delta \vec{x}_m + M \Delta \vec{x}_p = 0 \quad \dots(1)$$

The man walks displacement ($\Delta \vec{x}_{m/p} = -L\hat{i}$) relative to the plank. Denoting displacements of the man and the plank relative to the ground by $\Delta \vec{x}_m$ and $\Delta \vec{x}_p$, we can write

$$\Delta \vec{x}_{m/p} = \Delta \vec{x}_m - \Delta \vec{x}_p \rightarrow \Delta \vec{x}_m - \Delta \vec{x}_p = -L\hat{i} \quad \dots(2)$$

From the above equations (1) and (2), we have

$$\Delta \vec{x}_m = -\frac{ML\hat{i}}{m+M}$$

The man moves a distance $\frac{ML}{m+M}$ towards left relative to the ground.

$$\Delta \vec{x}_p = \frac{mL\hat{i}}{m+M}$$

The plank moves a distance $\frac{mL}{m+M}$ towards right relative to the ground.

3.4 Application of Methods of Work and Energy to a System of Particles

In a system of particles, all the particles occupy different locations at every instant of time and may change their locations with time. At an instant of time set of locations of all the particles of a system is known as configuration of the system. We say something has happened with the system only when some or all of its particles change their locations. It means that in every event or phenomena the system changes its configuration.

Methods of work and energy equips us to analyze what happens when a particle moves from one point of space to other. Now we will apply these methods to analyze a phenomenon in which a system of particle changes its configuration.

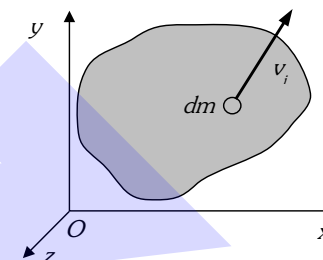
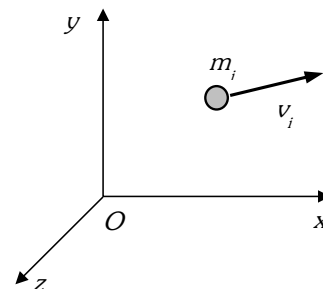
Kinetic Energy of a System of Particle

Kinetic energy of a system of particles is defined as sum of kinetic energies of all the particles of the system. If at an instant particles of masses $m_1, m_2, \dots, m_i, \dots, m_j, \dots, m_n$ are observed moving with velocities $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_i, \dots, \vec{v}_n$ respectively relative to a reference frame, the kinetic energy of the whole system relative to the reference frame is given by the following equation.

$$K = \frac{1}{2} \sum m_i v_i^2$$

If the system consists of continuous distribution of mass, instead of discrete particles, expression of kinetic energy becomes

$$K = \frac{1}{2} \int v^2 dm$$



Kinetic Energy of a System of Particle using Centroidal Frame

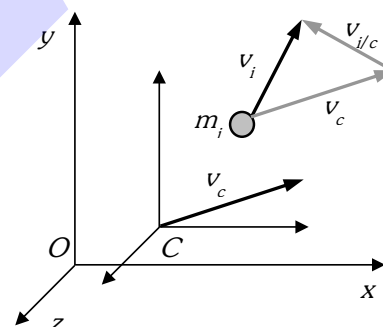
Centroidal frame of reference or center of mass frame is reference frame attached with the mass center of the system.

Let velocity of i^{th} particle of mass m_i is moving with velocity $\vec{v}_i$ relative to frame Oxyz. Mass center C and hence the centroidal frame Cxyz is moving with velocity $\vec{v}_c$. Therefore velocity of i^{th} particle relative to the centroidal frame is $\vec{v}_{i/c}$.

Kinetic energy of the whole system is given by the following equation.

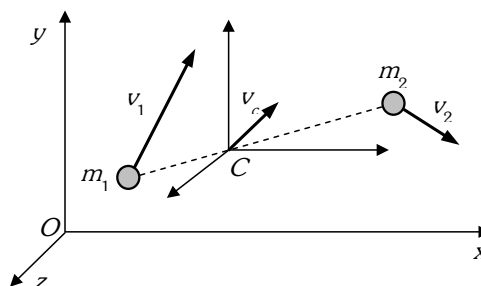
$$K = \frac{1}{2} \sum m_i v_i^2 = \frac{1}{2} \sum m_i v_c^2 + \frac{1}{2} \sum m_i v_{i/c}^2$$

Here the first term on the right hand side is kinetic energy due to translation of the mass center and the second term is kinetic energy of the system relative to the centroidal frame.



Kinetic Energy of a Two Particle System using Centroidal Frame

A two particle system consists of only two particles. Let a two particle system consists of particles of masses m_1 and m_2 moving with velocities $\vec{v}_1$ and $\vec{v}_2$ relative to a frame Oxyz. Their mass center C lies on the line joining them and divides separation between them in reciprocal ratio of masses m_1 and m_2 . The mass center and hence the centroidal frame is moving with velocity $\vec{v}_c$.



Kinetic energy of this two particle system relative to a frame O xyz is given by the following equation.

$$K = \frac{1}{2}(m_1 + m_2)v_c^2 + \frac{1}{2}\mu v_{rel}^2$$

The first term on the right hand side is kinetic energy due to translation of the mass center and the second term is kinetic energy of the system relative to the centroidal frame.

Here symbol μ is known as reduced mass of the two particle system and symbol v_{rel} is magnitude of velocity of either of the particles relative to the other.

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \quad \text{and} \quad v_{rel} = |\vec{v}_1 - \vec{v}_2| = |\vec{v}_2 - \vec{v}_1|$$

Work Energy Theorem for a System of Particles

The work energy theorem can be applied to each particle of the system. For i^{th} particle of the system, we can write

$$K_{i,i} + W_{i,i \rightarrow f} = K_{i,f}$$

Here $W_{i,i \rightarrow f}$ is total work done by all the internal forces $\vec{f}_{ij}$ and resultant external force $\vec{F}_i$ on the i^{th} particle, when the system goes from one configuration to other.

Adding kinetic energies of all particles, we can write kinetic energies K_i and K_f of the whole system in the initial as well as the final configuration. Adding work done $W_{i,i \rightarrow f}$ by internal as well as external forces on every particle we find total work done $W_{i \rightarrow f}$ by all the internal as well as external forces on the system. Now we can write work energy theorem.

$$K_i + W_{i \rightarrow f} = K_f$$

While applying the above equation to a system, care must be taken in calculating $W_{i \rightarrow f}$. In spite of the fact that the internal forces $\vec{f}_{ij}$ and $\vec{f}_{ji}$ being equal in magnitude and opposite in direction, the work done by them on the i^{th} and the j^{th} particles will not, in general, cancel out, since i^{th} and the j^{th} particles may undergo different amount of displacements.

The above description at first presents calculating of $W_{i \rightarrow f}$ as a cumbersome task. However for systems, which we usually encounter are not as complex as a general system of large number of particles may be. Systems which we usually face to analyze have limited number of particles or bodies interacting. For these systems we can simplify the task by calculating work of conservative internal forces as decrease in potential energy of the system. Total work of internal forces other than internal conservative forces vanishes, if these forces are due connecting inextensible links or links of constant length. These forces include string tension and normal reaction at direct contacts between the bodies included in the system. Work of internal forces of the kind other than these and work of external forces, can be calculated by definition of work.

Conservation of Mechanical Energy

If total work of internal forces other than conservative is zero and no external forces act on a system, total mechanical energy remains conserved.

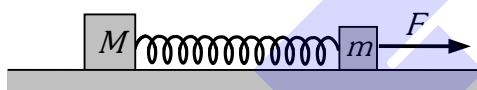
$$K_i + U_i = K_f + U_f$$

Since external forces are capable of changing mechanical energy of the system, under their presence total mechanical energy changes by amount equal to work $W_{\text{ext, i} \rightarrow \text{f}}$ done by all the external forces.

$$W_{\text{ext, i} \rightarrow \text{f}} = E_f - E_i = (K_f + U_f) - (K_i + U_i)$$

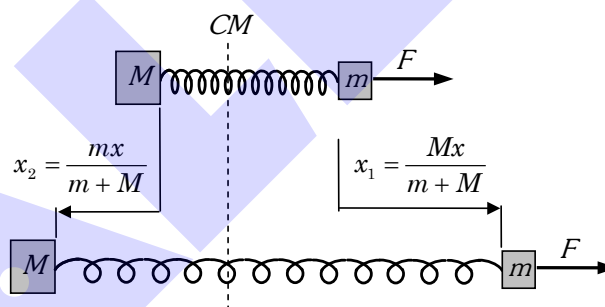
Illustrations

Illustration 24. Two blocks of masses m and M connected by a spring are placed on frictionless horizontal ground. When the spring is relaxed, a constant force F is applied as shown. Find maximum extension of the spring during subsequent motion.



Solution.

If we use ground as inertial frame as we usually do, solution of the problem becomes quite involved. Therefore, we prefer to use the centroidal frame, in which mass center remains at rest.



In the adjacent figure is shown horizontal position of mass center (CM) by dashed line. It remains unchanged in centroidal frame.

Mass center of two particle system divides separation between them in reciprocal ratio of the masses; therefore displacements x_1 and x_2 of the blocks must also be in reciprocal ratio of their masses. The extension x is sum of displacements x_1 and x_2 of the blocks as shown in the figure.

When extension of the spring achieves its maximum value, both the block must stop receding away from the mass center, therefore, velocities of both the blocks in centroidal frame must be zero.

During the process when spring is being extended, total work done by pseudo forces in centroidal frame become zero, negative work done by spring forces becomes equal to increase in potential energy and work done by the applied force evidently becomes Fx_1 .

Using above fact in applying work energy theorem on the system relative to the centroidal frame, we obtain

$$K_i + W_{i \rightarrow f} = K_f \rightarrow 0 + W_{i \rightarrow f, \text{spring force}} + W_{i \rightarrow f, F} = 0 \quad -\left(0 - \frac{1}{2} kx^2\right) + \frac{FMx}{m+m} = 0$$

$$x = \frac{2FM}{k(m+m)}$$

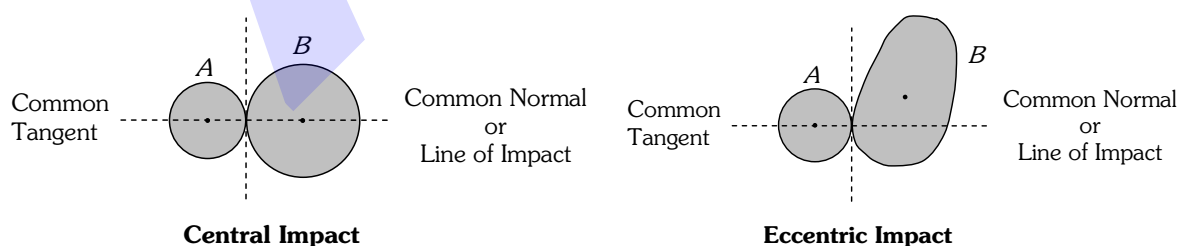
4.0 IMPACT BETWEEN TWO BODIES

Impact or collision is interaction of very small duration between two bodies in which the bodies apply relatively large forces on each other.

Interaction forces during an impact are created due to either direct contact or strong repulsive force fields or some connecting links. These forces are so large as compared to other external forces acting on either of the bodies that the effects of later can be neglected. The duration of the interaction is short enough as compared to the time scale of interest as to permit us only to consider the states of motion just before and after the event and not during the impact. Duration of an impact ranges from 10^{-23} s for impacts between elementary particles to millions of years for impacts between galaxies. The impacts we observe in our everyday life like that between two balls last from 10^{-3} s to few seconds.

4.1 Central and Eccentric Impact

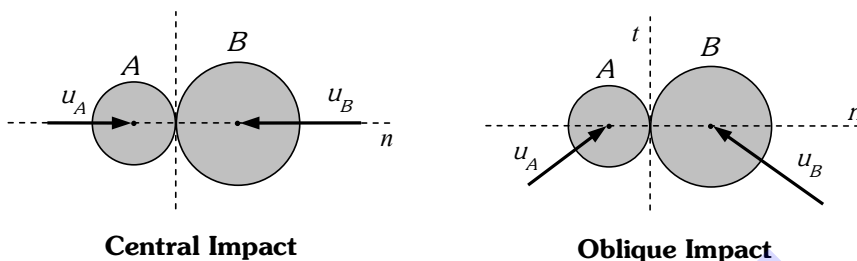
The common normal at the point of contact between the bodies is known as line of impact. If mass centers of the both the colliding bodies are located on the line of impact, the impact is called central impact and if mass centers of both or any one of the colliding bodies are not on the line of impact, the impact is called eccentric impact.



Central impact does not produce any rotation in either of the bodies whereas eccentric impact causes the body whose mass center is not on the line of impact to rotate. Therefore, at present we will discuss only central impact and postpone analysis of eccentric impact to cover after studying rotation motion.

Head-on (Direct) and Oblique Central Impact

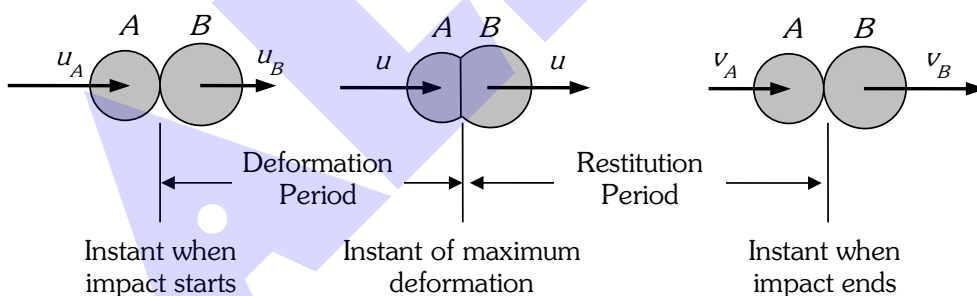
If velocities vectors of the colliding bodies are directed along the line of impact, the impact is called a direct or head-on impact; and if velocity vectors of both or of any one of the bodies are not along the line of impact, the impact is called an oblique impact.



In this chapter, we discuss only central impact, therefore the term central we usually not use and to these impacts, we call simply head-on and oblique impacts. Furthermore, use of the line of impact and the common tangent is so frequent in analysis of these impacts that we call them simply t-axis and n-axis.

4.2 Head-on (Direct) Central Impact

To understand what happens in a head-on impact let us consider two balls A and B of masses m_A and m_B moving with velocities u_A and u_B in the same direction as shown. Velocity u_A is larger than u_B so the ball A hits the ball B. During impact, both the bodies push each other and first they get deformed till the deformation reaches a maximum value and then they try to regain their original shape due to elastic behaviors of the materials forming the balls.



The time interval when deformation takes place is called the deformation period and the time interval in which the ball try to regain their original shapes is called the restitution period. Due to push applied by the balls on each other during period of deformation speed of the ball A decreases and that of the ball B increases and at the end of the deformation period, when the deformation is maximum both the ball move with the same velocity say it is u . Thereafter, the balls will either move together with this velocity or follow the period of restitution. During the period of restitution due to push applied by the balls on each other, speed of the ball A decrease further and that of ball B increase further till they separate from each other. Let us denote velocities of the balls A and B after the impact by v_A and v_B respectively.

Equation of Impulse and Momentum during impact

Impulse momentum principle describes motion of ball A during deformation period.

$$m_A u_A - \int D dt = m_A u \quad \dots(i)$$

Impulse momentum principle describes motion of ball B during deformation period.

$$m_B u_B + \int D dt = m_B u \quad \dots(ii)$$

Impulse momentum principle describes motion of ball A during restitution period.

$$m_A u - \int R dt = m_A v_A \quad \dots(iii)$$

Impulse momentum principle describes motion of ball B during restitution period.

$$m_B u + \int R dt = m_B v_B \quad \dots(iv)$$

Conservation of Momentum during impact

From equations, (i) and (ii) we have $m_A u_A + m_B u_B = (m_A + m_B) u \quad \dots(v)$

From equations, (iii) and (iv) we have $(m_A + m_B) u = m_A v_A + m_B v_B \quad \dots(vi)$

From equations, (v) and (vi) we obtain the following equation.

$$m_A v_A + m_B v_B = m_A u_A + m_B u_B \quad \dots(vii)$$

The above equation elucidates the principle of conservation of momentum.

Coefficient of Restitution

Usually the force D applied by the bodies A and B on each other during period differs from the force R applied by the bodies on each other during period of restitution. Therefore, it is not necessary that magnitude of impulse $\int D dt$ of deformation equals to the magnitude of impulse $\int R dt$ restitution.

The ratio of magnitudes of impulse of restitution to that of deformation is called the coefficient of restitution and is denoted by e .

$$e = \frac{\int R dt}{\int D dt} \quad \dots(viii)$$

Now from equations (i), (ii), (iii) and (iv), we have

$$e = \frac{v_B - v_A}{u_A - u_B} \quad \dots(ix)$$

Coefficient of restitution depend on various factors as elastic properties of materials forming the bodies, velocities of the contact points before impact, state of rotation of the bodies and temperature of the bodies. In general, its value ranges from zero to one but in collision where kinetic energy is generated its value may exceed one.

Depending on values of coefficient of restitution, two particular cases are of special interest.

Perfectly Plastic or Inelastic Impact : For these impacts $e = 0$, and bodies undergoing impact stick to each other after the impact.

Perfectly Elastic Impact : For these impacts $e = 1$.

Strategy to solve problems of head-on impact

Write momentum conservation equation

$$m_A v_A + m_B v_B = m_A u_A + m_B u_B \quad \dots(1)$$

Write rearranging terms of equation of coefficient of restitution

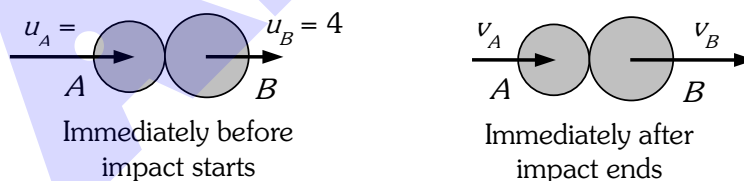
$$v_B - v_A = e(u_A - u_B) \quad \dots(2)$$

Use the above equations A and B.

Illustrations

Illustration 25. A ball of mass 2 kg moving with speed 5 m/s collides directly with another of mass 3 kg moving in the same direction with speed 4 m/s. The coefficient of restitution is $\frac{2}{3}$. Find the velocities after collision.

Solution. Denoting the first ball by A and the second ball by B velocities immediately before and after the impact are shown in the figure.



Applying principle of conservation of momentum, we have

$$m_B v_B + m_A v_A = m_A u_A + m_B u_B \rightarrow 3v_B + 2v_A = 2 \times 5 + 3 \times 4$$

$$3v_B + 2v_A = 22 \quad \dots(i)$$

Applying equation of coefficient of restitution, we have

$$v_B - v_A = e(u_A - u_B) \rightarrow v_B - v_A = \frac{2}{3}(5 - 4)$$

$$3v_B - 3v_A = 2 \quad \dots(ii)$$

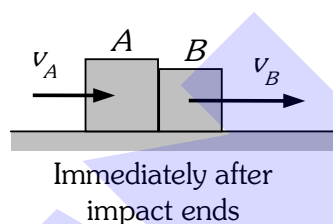
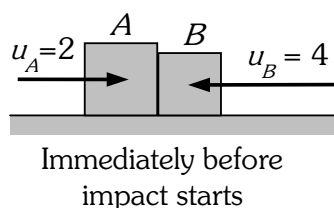
From equation (i) and (ii), we have $v_A = 4$ m/s and $v_B = 4.67$ m/s **Ans.**

Illustration 26. A block of mass 5 kg moves from left to right with a velocity of 2 m/s and collides with another block of mass 3 kg moving along the same line in the opposite direction with velocity 4 m/s.

- (a) If the collision is perfectly elastic, determine velocities of both the blocks after their collision.
- (b) If coefficient of restitution is 0.6, determine velocities of both the blocks after their collision.

Solution.

Denoting the first block by A and the second block by B velocities immediately before and after the impact are shown in the figure.



Applying principle of conservation of momentum, we have

$$m_B v_B + m_A v_A = m_A u_A + m_B u_B \rightarrow 3v_B + 5v_A = 5 \times 2 + 3 \times (-4)$$

$$3v_B + 5v_A = -2 \quad \dots(i)$$

Applying equation of coefficient of restitution, we have

$$v_B - v_A = e(u_A - u_B) \rightarrow v_B - v_A = e\{2 - (-4)\}$$

$$v_B - v_A = 6e \quad \dots(ii)$$

- (a) For perfectly elastic impact $e = 1$. Using this value in equation (ii), we have

$$v_B - v_A = 6 \quad \dots(iia)$$

Now from equation (i) and (iia), we obtain

$$v_A = -2.5 \text{ m/s and } v_B = 3.5 \text{ m/s}$$

- (b) For value $e = 0.6$, equation 2 is modified as

$$v_B - v_A = 3.6 \quad (iib)$$

Now from equation (i) and (iib), we obtain

$$v_A = -1.6 \text{ m/s and } v_B = 2.0 \text{ m/s}$$

Block A reverse back with speed 1.6 m/s and B also move in opposite direction to its original direction with speed 2.0 m/s.

Illustration 27. Two identical balls A and B moving with velocities u_A and u_B in the same direction collide. Coefficient of restitution is e .

- (a) Deduce expression for velocities of the balls after the collision.
- (b) If collision is perfectly elastic, what do you observe?

Solution.

Equation expressing momentum conservation is

$$v_A + v_B = u_A + u_B \quad \dots(A)$$

Equation of coefficient of restitution is

$$v_B - v_A = eu_A - eu_B \quad \dots(B)$$

(a) From the above two equations, velocities v_A and v_B are

$$v_A = \left(\frac{1-e}{2}\right)u_A + \left(\frac{1+e}{2}\right)u_B \quad \dots(i)$$

$$v_B = \left(\frac{1+e}{2}\right)u_A + \left(\frac{1-e}{2}\right)u_B \quad \dots(ii)$$

(b) For perfectly elastic impact $e = 1$, velocities v_A and v_B are

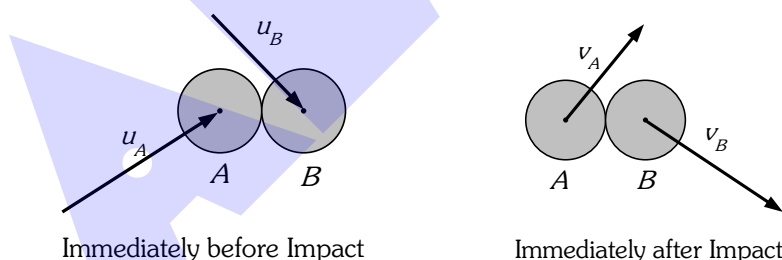
$$v_A = u_B \quad \dots(iii)$$

$$v_B = u_A \quad \dots(iv)$$

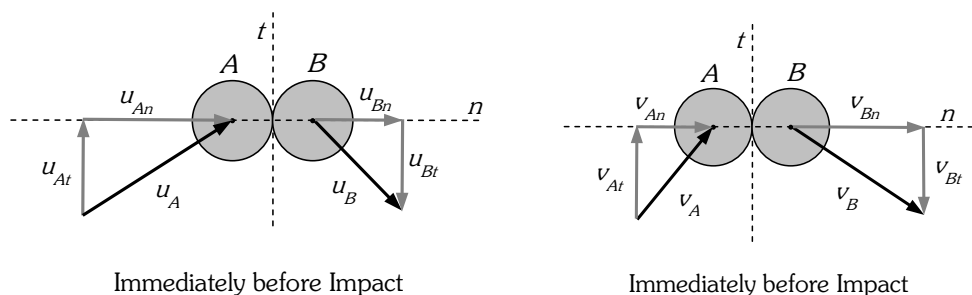
Identical bodies exchange their velocities after perfectly elastic impact.

4.3 Oblique Central Impact

In oblique central impact, velocity vectors of both or of any one of the bodies are not along the line of impact and mass center of bodies are on the line of impact. Due to impact speeds and direction of motion of both the balls change. In the given figure is shown two balls A and B of masses m_A and m_B moving with velocities u_A and u_B collide obliquely. After the collision let they move with velocities v_A and v_B as shown in the next figure.



To analyze the impact, we show components of velocities before and after the impact along the common tangent and the line of impact. These components are shown in the following figure.



Component along the t-axis If surfaces of the bodies undergoing impact are smooth, they cannot apply any force on each other along the t-axis and component of momentum along the t-axis of each bodies, considered separately, is conserved. Hence, t-component of velocities of each of the bodies remains unchanged.

$$v_{At} = u_{At} \text{ and } v_{Bt} = u_{Bt} \quad \dots(A)$$

Component along the n-axis For components of velocities along the n-axis, the impact can be treated same as head-on central impact.

The component along the n-axis of the total momentum of the two bodies is conserved

$$m_B v_{Bn} + m_A v_{An} = m_B u_{Bn} + m_A u_{An} \quad \dots(B)$$

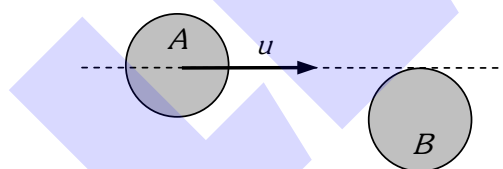
Concept of coefficient of restitution e is applicable only for the n-component velocities.

$$v_{Bn} - v_{An} = e(u_{An} - u_{Bn}) \quad \dots(C)$$

The above four independent equation can be used to analyze oblique central impact of two freely moving bodies.

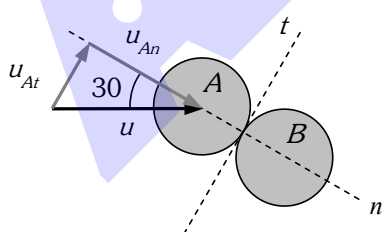
Illustrations

Illustration 28. A disk sliding with velocity u on a smooth horizontal plane strikes another identical disk kept at rest as shown in the figure. If the impact between the disks is perfectly elastic impact, find velocities of the disks after the impact.

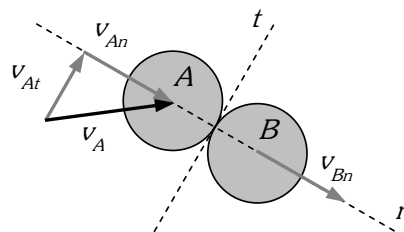


Solution.

- (a) We first show velocity components along the t and the n-axis immediately before and after the impact. angle that the line of impact makes with velocity u is 30° .



Immediately before Impact



Immediately after Impact

Component along t-axis Components of momentum along the t-axis of each disk, considered separately, is conserved. Hence, t-component of velocities of each of the bodies remains unchanged.

$$v_{At} = u_{At} = \frac{u}{2} \text{ and } v_{Bt} = u_{Bt} = 0 \quad \dots(i)$$

Component along n-axis. The component along the n-axis of the total momentum of the two bodies is conserved

$$m_B v_{Bn} + m_A v_{An} = m_B u_{Bn} + m_A u_{An} \rightarrow m v_{Bn} + m v_{An} = m \times 0 + m \frac{u\sqrt{3}}{2}$$

$$v_{Bn} + v_{An} = \frac{u\sqrt{3}}{2} \quad \dots(ii)$$

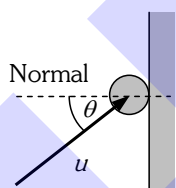
Concept of coefficient of restitution e is applicable only for the n-component velocities.

$$v_{Bn} - v_{An} = e(u_{An} - u_{Bn}) \rightarrow v_{Bn} - v_{An} = \frac{u\sqrt{3}}{2} \quad \dots(iii)$$

$$\text{From equations (ii) and (iii), we have } v_{An} = 0 \text{ and } v_{Bn} = \frac{u\sqrt{3}}{2} \quad \dots(iv)$$

From equations (i) and (iv) we can write velocities of both the disks.

Illustration 29. A ball collides with a frictionless wall with velocity u as shown in the figure. Coefficient of restitution for the impact is e .

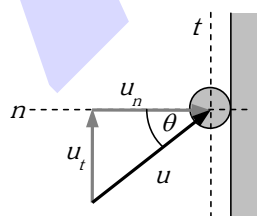


- (a) Find expression for the velocity of the ball immediately after the impact.
 (b) If impact is perfectly elastic what do you observe?

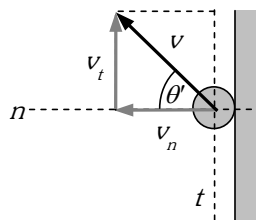
Solution.

- (a) Let us consider the ball as the body A and the wall as the body B. Since the wall has infinitely large inertia (mass) as compared to the ball, the state of motion of the wall, remains unaltered during the impact i.e. the wall remain stationary.

Now we show velocities of the ball and its t and n-components immediately before and after the impact. For the purpose we have assumed velocity of the ball after the impact v .



Immediately before Impact



Immediately after Impact

Component along t-axis : Components of momentum along the t-axis of the ball is conserved. Hence, t-component of velocities of each of the bodies remains unchanged.

$$v_t = u_t = u \sin \theta \quad \dots(i)$$

Component along n-axis : Concept of coefficient of restitution e is applicable only for the n-component velocities.

$$v_{Bn} - v_{An} = e(u_{An} - u_{Bn}) \rightarrow -v_n = eu_n$$

$$v_n = -eu \cos \theta \quad \dots(ii)$$

From equations (i) and (ii), the t and n-components of velocity of the ball after the impact are

$$v_t = u \sin \theta \quad \text{and} \quad v_n = eu \sin \theta$$

(b) If the impact is perfectly elastic, we have $v_t = u \sin \theta$, $v_n = eu \sin \theta$ and $\theta' = \theta$

The ball will rebound with the same speed making the same angle with the vertical at which it has collided. In other words, a perfectly elastic collision of a ball with a wall follows the same laws as light follows in reflection at a plane mirror.

5.0 VARIABLE MASS SYSTEM

If a mass is added or ejected from a system, at rate μ kg/s and relative velocity $\vec{v}_{rel}$ (w.r.t. the system), then the force exerted by this mass on the system has magnitude $\mu |\vec{v}_{rel}|$.

Thrust Force ($\vec{F}_t$)

$$\vec{F}_t = \vec{v}_{rel} \left(\frac{dm}{dt} \right)$$

Suppose at some moment $t = t$ mass of a body is m and its velocity is $\vec{v}$. After some time at $t = t + dt$ its mass becomes $(m - dm)$ and velocity becomes $\vec{v} + d\vec{v}$. The mass dm is ejected with relative velocity $\vec{v}_r$. Absolute velocity of mass ' dm ' is therefore $(\vec{v} + \vec{v}_r)$. If no external forces are acting on the system, the linear momentum of the system will remain conserved, or

$$\vec{P}_i = \vec{P}_f$$

$$\text{or} \quad m\vec{v} = (m - dm)(\vec{v} + d\vec{v}) + dm(\vec{v} + \vec{v}_r)$$

$$\text{or} \quad m\vec{v} = m\vec{v} + md\vec{v} - (dm)\vec{v} - (dm)(d\vec{v}) + (dm)\vec{v} + \vec{v}_r dm$$

The term $(dm)(d\vec{v})$ is too small and can be neglected.

$$\therefore \quad md\vec{v} = -\vec{v}_r dm$$

$$\text{or} \quad m \left(\frac{d\vec{v}}{dt} \right) = \vec{v}_r \left(-\frac{dm}{dt} \right)$$

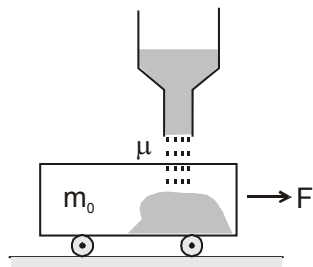
$$\text{Here, } m \left(-\frac{d\vec{v}}{dt} \right) = \text{thrust force } (\vec{F}_t)$$

and $-\frac{dm}{dt}$ = rate at which mass is ejecting

$$\text{or} \quad \vec{F}_t = \vec{v}_r \left(\frac{dm}{dt} \right)$$

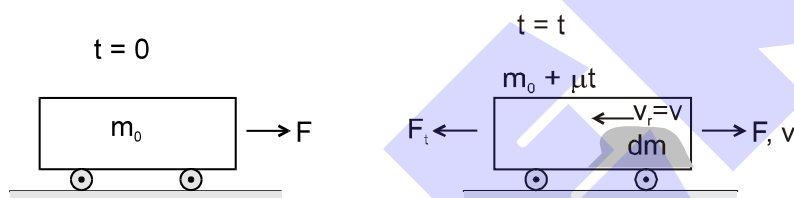
Illustrations

Illustration 30. A flat car of mass m_0 starts moving to the right due to a constant horizontal force F . Sand spills on the flat car from a stationary hopper. The rate of loading is constant and equal to μ kg/s. Find the time dependence of the velocity and the acceleration of the flat car in the process of loading. The friction is negligibly small.



Solution

Initial velocity of the flat car is zero. Let v be its velocity at time t and m its mass at that instant. Then



At $t = 0$, $v = 0$ and $m = m_0$ at $t = t$, $v = v$ and $m = m_0 + \mu t$

Here, $v_r = v$ (backwards)

$$\frac{dm}{dt} = \mu$$

$$\therefore F_t = v_r \frac{dm}{dt} = \mu v \quad (\text{backwards})$$

Net force on the flat car at time t is $F_{\text{net}} = F - F_t$

$$\text{or } m \frac{dv}{dt} = F - \mu v \quad \dots(i)$$

$$\text{or } (m_0 + \mu t) \frac{dv}{dt} = F - \mu v$$

$$\text{or } \int_0^v \frac{dv}{F - \mu v} = \int_0^t \frac{dt}{m_0 + \mu t}$$

$$\therefore -\frac{1}{\mu} [\ln(F - \mu v)]_0^v = \frac{1}{\mu} [\ln(m_0 + \mu t)]_0^t$$

$$\Rightarrow \ln\left(\frac{F}{F - \mu v}\right) = \ln\left(\frac{m_0 + \mu t}{m_0}\right)$$

$$\therefore \frac{F}{F - \mu v} = \frac{m_0 + \mu t}{m_0}$$

$$\text{or } v = \frac{Ft}{m_0 + \mu t} \quad \text{Ans.}$$

From Eq. (i), $\frac{dv}{dt}$ = acceleration of flat car at time t

$$\text{or } = \frac{F - \mu v}{m}$$

$$a = \left(\frac{F - \frac{F\mu t}{m_0 + \mu t}}{m_0 + \mu t} \right) \text{ or } a = \frac{Fm_0}{(m_0 + \mu t)^2} \quad \text{Ans.}$$

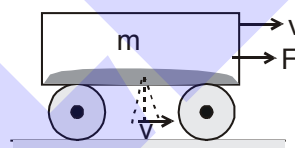
Illustration 31. A cart loaded with sand moves along a horizontal floor due to a constant force F coinciding in direction with the cart's velocity vector. In the process sand spills through a hole in the bottom with a constant rate $\mu \text{ kg/s}$. Find the acceleration and velocity of the cart at the moment t , if at the initial moment $t = 0$ the cart with loaded sand had the mass m_0 and its velocity was equal to zero. Friction is to be neglected.

Solution

In this problem the sand spills through a hole in the bottom of the cart. Hence, the relative velocity of the sand v_r will be zero because it will acquire the same velocity as that of the cart at the moment.

$$v_r = 0$$

$$\text{Thus, } F_t = 0 \quad \left(\text{as } F_t = v_r \frac{dm}{dt} \right)$$



and the net force will be F only.

$$\therefore F_{\text{net}} = F$$

$$\text{or } m \left(\frac{dv}{dt} \right) = F \quad \dots(i)$$

$$\text{But here } m = m_0 - \mu t$$

$$\therefore (m_0 - \mu t) \frac{dv}{dt} = F$$

$$\text{or } \int_0^v dv = \int_0^t \frac{F dt}{m_0 - \mu t}$$

$$\therefore v = \frac{F}{-\mu} \left[\ln(m_0 - \mu t) \right]_0^t$$

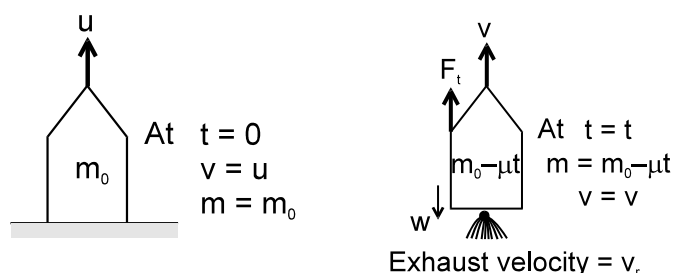
$$\text{or } v = \frac{F}{\mu} \ln \left(\frac{m_0}{m_0 - \mu t} \right)$$

From eq. (i), acceleration of the cart

$$a = \frac{dv}{dt} = \frac{F}{m} \quad \text{or} \quad a = \frac{F}{m_0 - \mu t}$$

5.1 Rocket propulsion

Let m_0 be the mass of the rocket at time $t = 0$. m its mass at any time t and v its velocity at that moment. Initially, let us suppose that the velocity of the rocket is u .



Further, let $\left(\frac{-dm}{dt}\right)$ be the mass of the gas ejected per unit time and v_r the exhaust velocity of the gases with respect to rocket. Usually $\left(\frac{-dm}{dt}\right)$ and v_r are kept constant throughout the journey of the rocket. Now, let us write few equations which can be used in the problems of rocket propulsion. At time $t = t$,

1. Thrust force on the rocket $F_t = v_r \left(\frac{-dm}{dt}\right)$ (upwards)
 2. Weight of the rocket $W = mg$ (downwards)
 3. Net force on the rocket $F_{\text{net}} = F_t - W$ (upwards)
- or $F_{\text{net}} = v_r \left(\frac{-dm}{dt}\right) - mg$
4. Net acceleration of the rocket $a = \frac{F}{m}$

or $\frac{dv}{dt} = \frac{v_r}{m} \left(\frac{-dm}{dt}\right) - g$

or $dv = \frac{v_r}{m} (-dm) - g dt$

or $\int_u^v dv = v_r \int_{m_0}^m \frac{-dm}{m} - g \int_0^t dt$

Thus, $v = u - gt + v_r \ln\left(\frac{m_0}{m}\right)$... (i)

Note : 1. $F_t = v_r \left(\frac{-dm}{dt}\right)$ is upwards, as v_r is downwards and $\frac{dm}{dt}$ is negative.

2. If gravity is ignored and initial velocity of the rocket $u = 0$, Eq. (i) reduces to $v = v_r \ln\left(\frac{m_0}{m}\right)$.

Illustrations

Illustration 32. A rocket, with an initial mass of 1000 kg, is launched vertically upwards from rest under gravity. The rocket burns fuel at the rate of 10 kg per second. The burnt matter is ejected vertically downwards with a speed of 2000 ms^{-1} relative to the rocket. If burning stops after one minute. Find the maximum velocity of the rocket. (Take g as 10 ms^{-2})

Solution Using the velocity equation

$$v = u - gt + v_r \ln\left(\frac{m_0}{m}\right)$$

Here $u = 0$, $t = 60\text{s}$, $g = 10 \text{ m/s}^2$, $v_r = 2000 \text{ m/s}$, $m_0 = 1000 \text{ kg}$
and $m = 1000 - 10 \times 60 = 400 \text{ kg}$

We get
$$v = 0 - 600 + 2000 \ln\left(\frac{1000}{400}\right)$$

or
$$v = 2000 \ln 2.5 - 600$$

The maximum velocity of the rocket is $200(10 \ln 2.5 - 3) = 1232.6 \text{ ms}^{-1}$ **Ans.**

Illustration 33. A uniform chain of mass m and length ℓ hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the table on the chain when half of its length has fallen on the table. The fallen part does not form heap.

Solution 1. Weight of the portion BC of the chain

lying on the table, $W = \frac{mg}{2}$ (downwards) Using $v = \sqrt{2gh}$

2. Thrust force $F_t = v_r \left(\frac{dm}{dt}\right)$

$$v_r = v$$

$$\frac{dm}{dt} = \lambda v$$

$$F_t = \lambda v^2 \quad (\text{where, } \lambda = m/\ell, \text{ is mass per unit length of chain})$$

$$v^2 = (\sqrt{g\ell})^2 = g\ell$$

$$\therefore F_t = \left(\frac{m}{\ell}\right)(g\ell) = mg \quad (\text{downwards})$$

$\therefore$ Net force exerted by the chain on the table is

$$F = W + F_t = \frac{mg}{2} + mg = \frac{3}{2}mg$$

So, from Newton's third law the force exerted by the table on the chain will be $\frac{3}{2}mg$ (vertically upwards).

